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Machine Learning Project Report

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Module: Introfuction to Statistical
Learning

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Abstract

Our project applies regression and classification pipeline to two biological datasets with very different characteristics, a large protein dataset with 45,730 observations and a small clinical dataset with 270 patients.

On the clinical dataset, we trained five classification models to predict heart disease presence. Logistic Regression achieved 84.0% accuracy and an AUC of 0.922, correctly identifying 80.6% of true heart disease cases.

On the protein dataset, we used three variable selection methods to predict hydrophobic surface exposure. All three methods independently selected the same six variables. Adding a polynomial term after residual diagnostics raised the test R^2 from 0.681 to 0.775.

On the same clinical dataset, we then applied the full regression pipeline to predict maximum heart rate. Every model failed, test R^2 never exceeded 0.037. This was not a mistake. It demonstrates what happens when the same rigorous methodology is applied to a dataset that is too small and too noisy for regression to generalise.

The contrast between a test R^2 of 0.775 and a test R^2 of 0.037, produced by the same pipeline on two different datasets, is the central finding of our project.

1 Data Description and Research Objectives

This project evaluates machine learning methodologies across two fundamentally different biological environments: a highly deterministic, geometric simulation and a noisy, real-world clinical setting.

1.1 Dataset 1: Physicochemical Properties of Proteins

Source of the Data

This dataset contains 45,730 observations of protein tertiary structures. It is publicly sourced from the UCI Machine Learning Repository and can be accessed here: [Physicochemical Properties of Protein Tertiary Structure](#).

Description of Variables

The dataset consists of 10 continuous variables measuring the physical 3D geometry of the folded proteins:

- **RMSD:** Size of the residue.
- **F1:** Total surface area.
- **F2:** Non polar exposed area.
- **F3:** Fractional area of exposed non polar residue.
- **F4:** Fractional area of exposed non polar part of residue.
- **F5:** Molecular mass weighted exposed area.
- **F6:** Average deviation from standard exposed area of residue.
- **F7:** Euclidian distance.
- **F8:** Secondary structure penalty.
- **F9:** Spacial Distribution constraints (N,K Value).

Explanation of the Research Question

Can we predict the hydrophobic surface exposure (F3) of a protein using its other physicochemical properties? Because this dataset represents a closed system governed by strict 3D geometric laws, our primary research objective is to test how algorithms handle severe multicollinearity and whether they can successfully capture non-linear, deterministic physical boundaries.

1.2 Dataset 2: Clinical Heart Disease Diagnostics

Source of the Data

This dataset contains cardiovascular measurements and demographic information for 270 human patients. It is publicly sourced and can be accessed here: [Heart Disease Prediction Dataset](#).

Description of Variables

The dataset contains 14 clinical attributes encompassing both categorical and continuous data:

- **Age:** The patient's age in years.
- **Sex:** Biological sex. 1 = male, 0 = female.
- **Chest pain type:** Type of chest pain reported. 1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic.
- **BP:** Resting blood pressure measured in mmHg at hospital admission.
- **Cholesterol:** Serum cholesterol level in mg/dL.
- **FBS over 120:** Fasting blood sugar. 1 = blood sugar above 120 mg/dL, 0 = below. Used as a basic diabetes indicator.
- **EKG results:** Resting electrocardiogram results. 0 = normal, 1 = ST-T wave abnormality, 2 = left ventricular hypertrophy.
- **Max HR:** Maximum heart rate achieved during an exercise stress test, measured in beats per minute.
- **Exercise angina:** Whether the patient experienced chest pain during exercise. 1 = yes, 0 = no.
- **ST depression:** Depression in the ST segment of the ECG induced by exercise relative to rest. Higher values indicate more cardiac stress.
- **Slope of ST:** The slope of the peak exercise ST segment. 1 = upsloping, 2 = flat, 3 = downsloping.
- **Number of vessels fluoro:** Number of major coronary vessels visible under fluoroscopy (0 to 3). More visible vessels generally indicate more blockage.
- **Thallium:** Result of a thallium stress test measuring blood flow to the heart. 3 = normal, 6 = fixed defect, 7 = reversible defect.
- **Heart Disease:** The target variable. Presence or Absence of heart disease diagnosis.

Explanation of the Research Question

Can machine learning algorithms autonomously discover established physiological rules and provide accurate clinical diagnostics from highly variable human data? This dataset tests two objectives:

1. *Diagnostic Classification:* Can classical algorithms accurately classify heart disease presence while prioritizing medical sensitivity?
2. *Physiological Regression:* Can regression autonomously derive the universally established linear decline in maximum heart rate ($220 - \text{Age}$), despite the high variance inherent to human biology?

Classification Analysis : Heart Disease Dataset

Target: Heart Disease (Presence / Absence)

Question: Can we predict whether a patient has heart disease from their clinical measurements?

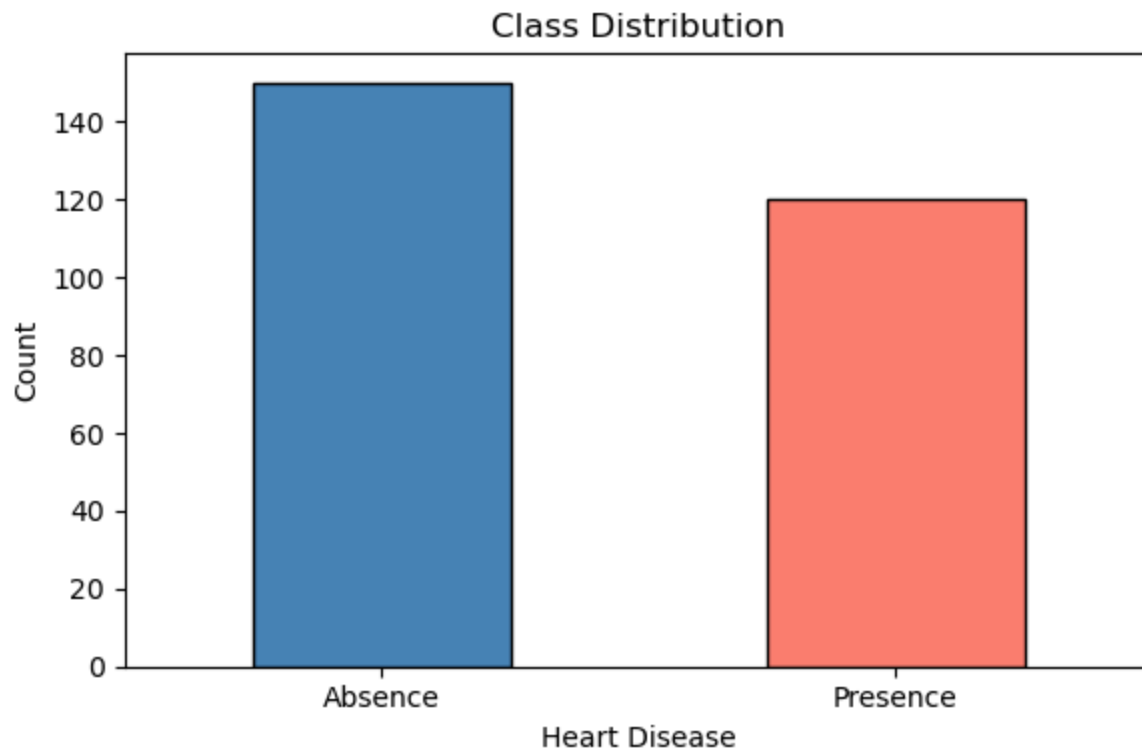
We load the dataset and explore its basic structure before any modelling.

```
Target Distribution:
Heart Disease
Absence      150
Presence     120
Name: count, dtype: int64
Heart Disease
Absence      0.556
Presence     0.444
Name: proportion, dtype: float64
```

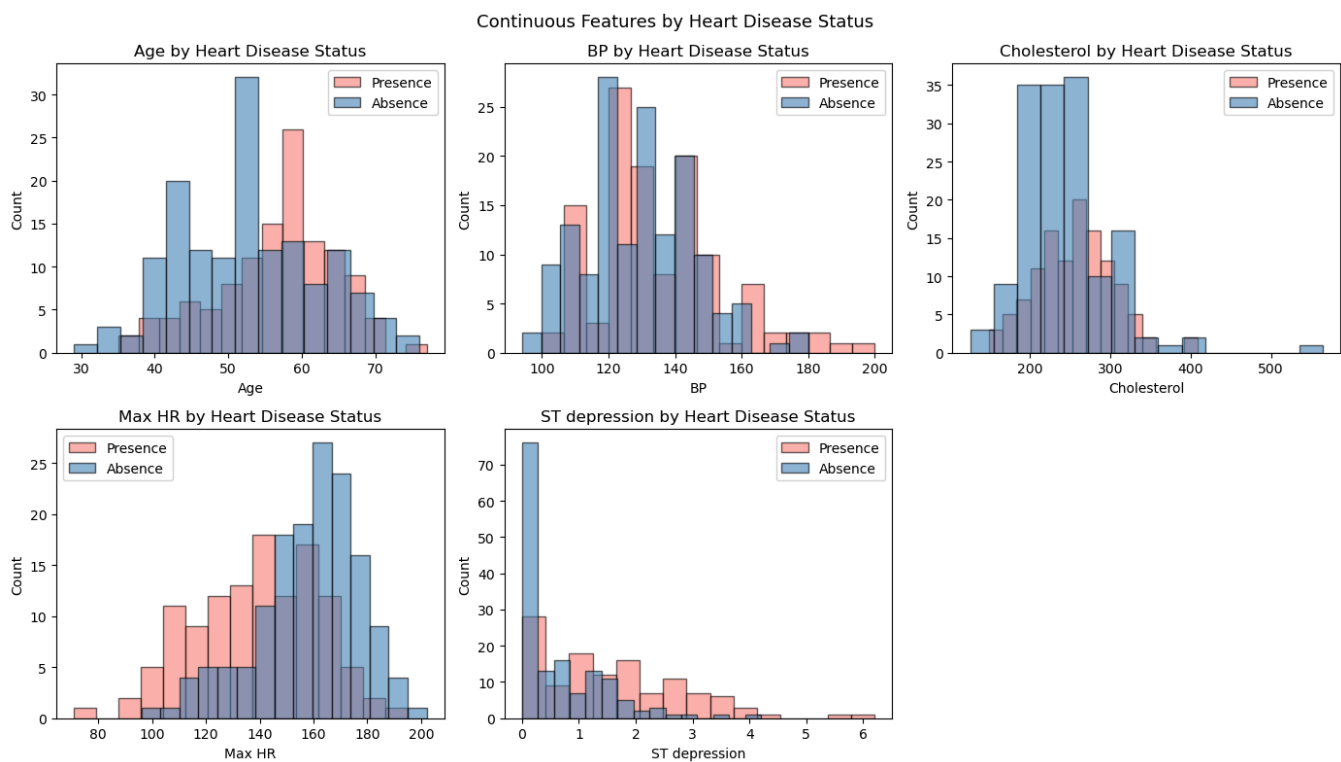
270 patients, 14 variables, no missing values. The target is binary: 150 patients without heart disease (55.6%) and 120 with it (44.4%). The dataset is reasonably balanced.

1. Exploratory Data Analysis

We visualise continuous and categorical features grouped by heart disease status to understand which variables differ between the two groups.

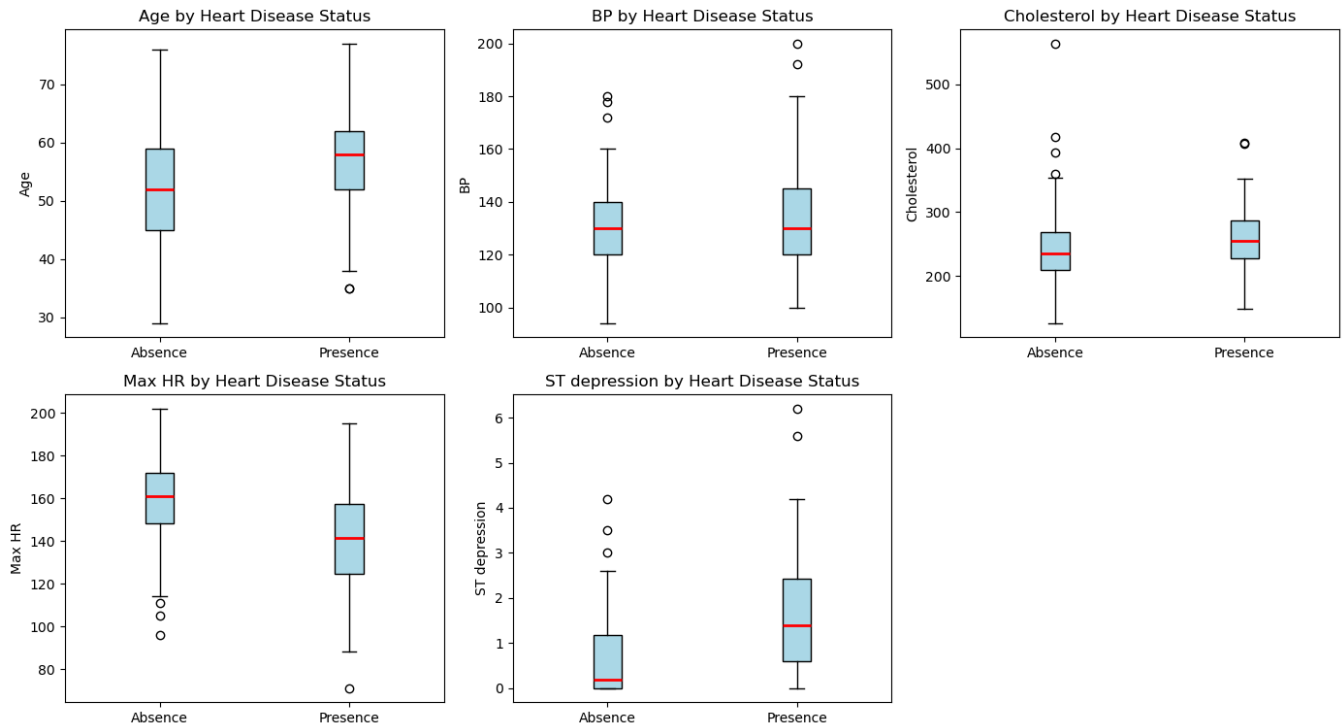


The dataset is nearly balanced : 150 absence vs 120 presence cases. No resampling needed.



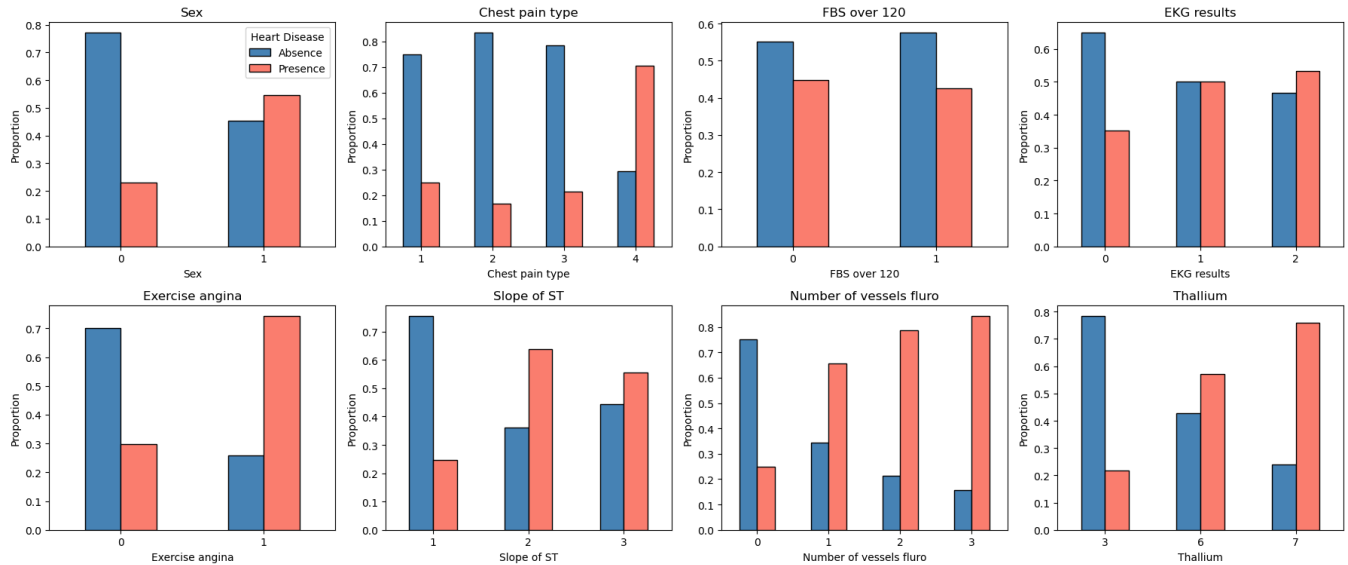
Patients with heart disease tend to have lower maximum heart rate and higher ST depression compared to healthy patients.

Boxplots of Continuous Features by Heart Disease Status



The boxplots confirm the histogram findings. Maximum heart rate is clearly lower in the presence group. ST depression is higher and more variable.

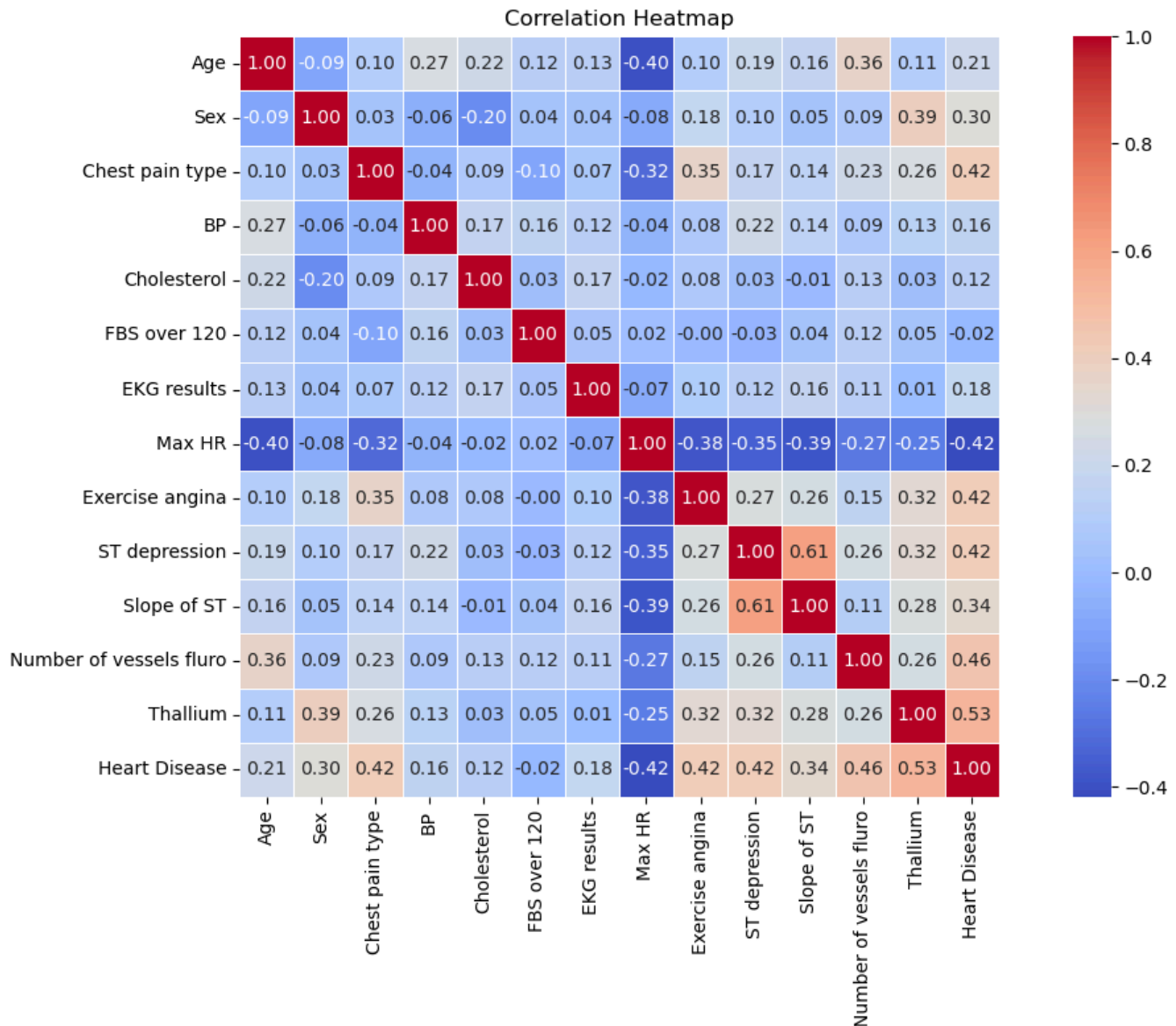
Categorical Features vs Heart Disease



Chest pain type 4 and higher numbers of fluoroscopy vessels are strongly associated with heart disease presence. Sex also shows a notable difference between the two groups.

2. Correlation Analysis

We examine correlations between all variables to understand their relationships before modelling.



Maximum heart rate is negatively correlated with most cardiac stress indicators. Number of vessels and ST depression show the strongest positive correlations with heart disease. No pair of predictors is perfectly correlated so no variable needs to be removed.

3. Outlier Detection

We apply the IQR method to identify extreme values in the continuous variables.

Outlier Detection (IQR Method):

Age: 0 outliers (lower bound=28.5, upper bound=80.5)

BP: 9 outliers (lower bound=90.0, upper bound=170.0)

Cholesterol: 5 outliers (lower bound=112.5, upper bound=380.5)

Max HR: 1 outliers (lower bound=83.5, upper bound=215.5)

ST depression: 4 outliers (lower bound=-2.4, upper bound=4.0)

A small number of outliers detected, at most 9 for blood pressure. We retain all observations given the small dataset size of 270 patients.

4. Preprocessing

We encode the target as binary, apply one-hot encoding to categorical predictors, and prepare the feature matrix for modelling.

Final feature matrix shape: (270, 20)

Features: ['Age', 'Sex', 'BP', 'Cholesterol', 'FBS over 120', 'Max HR', 'Exercise angina', 'S T depression', 'Chest pain type_2', 'Chest pain type_3', 'Chest pain type_4', 'EKG results_1', 'EKG results_2', 'Slope of ST_2', 'Slope of ST_3', 'Number of vessels fluro_1', 'Number of vessels fluro_2', 'Number of vessels fluro_3', 'Thallium_6', 'Thallium_7']

After encoding, the feature matrix contains 20 columns. The first category of each categorical variable is dropped to avoid multicollinearity.

5. Baseline Logistic Regression on Full Dataset

Before splitting the data we fit a logistic regression on the full dataset to identify which variables are statistically significant predictors of heart disease.

Optimization terminated successfully.

Current function value: 0.299370

Iterations 8

Logit Regression Results

```

=====
Dep. Variable:          Target    No. Observations:          270
Model:                  Logit     Df Residuals:              249
Method:                  MLE      Df Model:                  20
Date:                   Thu, 19 Mar 2026    Pseudo R-squ.:            0.5642
Time:                   19:39:39    Log-Likelihood:            -80.830
converged:              True      LL-Null:                   -185.48
Covariance Type:        nonrobust    LLR p-value:               1.613e-33
=====
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```

	coef	std err	z	P> z	[0.02
5 0.975]					

Intercept	-7.6870	3.288	-2.338	0.019	-14.13
1 -1.243					
Q("Chest pain type_2")[T.True]	1.7412	0.946	1.841	0.066	-0.11
3 3.595					
Q("Chest pain type_3")[T.True]	0.7849	0.792	0.991	0.322	-0.76
8 2.338					
Q("Chest pain type_4")[T.True]	2.7487	0.813	3.381	0.001	1.15
5 4.342					
Q("EKG results_1")[T.True]	0.8036	3.562	0.226	0.822	-6.17
8 7.785					
Q("EKG results_2")[T.True]	0.6768	0.426	1.590	0.112	-0.15
8 1.511					
Q("Slope of ST_2")[T.True]	1.1131	0.514	2.164	0.030	0.10
5 2.121					
Q("Slope of ST_3")[T.True]	0.1284	1.061	0.121	0.904	-1.95
2 2.209					
Q("Number of vessels fluro_1")[T.True]	2.1521	0.560	3.845	0.000	1.05
5 3.249					
Q("Number of vessels fluro_2")[T.True]	3.1005	0.808	3.839	0.000	1.51
8 4.683					
Q("Number of vessels fluro_3")[T.True]	2.1647	0.926	2.337	0.019	0.35
0 3.980					
Thallium_6[T.True]	-0.3189	0.865	-0.369	0.712	-2.01
5 1.377					
Thallium_7[T.True]	1.4687	0.465	3.156	0.002	0.55
7 2.381					
Age	-0.0251	0.027	-0.944	0.345	-0.07
7 0.027					
Sex	1.8990	0.606	3.133	0.002	0.71
1 3.087					
BP	0.0311	0.013	2.431	0.015	0.00
6 0.056					
Cholesterol	0.0066	0.004	1.525	0.127	-0.00
2 0.015					
Q("FBS over 120")	-0.3760	0.607	-0.620	0.536	-1.56
6 0.814					
Q("Max HR")	-0.0205	0.012	-1.681	0.093	-0.04

```

4      0.003
Q("Exercise angina")      0.5347      0.468      1.143      0.253      -0.38
2      1.451
Q("ST depression")      0.4761      0.253      1.883      0.060      -0.01
9      0.972

```

```

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```

Significant Predictors (p < 0.05):

```

Q("Number of vessels fluro_1")[T.True]      0.0001
Q("Number of vessels fluro_2")[T.True]      0.0001
Q("Chest pain type_4")[T.True]              0.0007
Thallium_7[T.True]                          0.0016
Sex                                           0.0017
BP                                           0.0151
Q("Number of vessels fluro_3")[T.True]      0.0194
Q("Slope of ST_2")[T.True]                  0.0305
dtype: float64

```

Pseudo $R^2 = 0.564$, indicating a good fit. The most significant predictors are number of fluoroscopy vessels, chest pain type 4, sex, blood pressure, thallium score, and slope of ST.

6. Train / Test Split (70% / 30%)

We split the data with stratification to preserve class balance. Continuous variables are standardised after splitting to prevent data leakage.

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Training set : 189 samples (70%)
Test set     : 81 samples (30%)

```

```

Class balance in train: {0: 105, 1: 84}
Class balance in test : {0: 45, 1: 36}

```

Training set: 189 patients. Test set: 81 patients. Both sets maintain a similar class ratio. The scaler is fitted on training data only and applied to the test set to prevent data leakage.

7. Model Training and Evaluation

We define a reusable evaluation function and train five classification models. Each is evaluated using accuracy, sensitivity, specificity, and AUC.

Logistic Regression

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Logistic Regression
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```

Confusion Matrix:

	Predicted Absence	Predicted Presence
Actual Absence	39	6
Actual Presence	7	29

Accuracy : 0.8395 (84.0%)
 Sensitivity : 0.8056 (80.6%)
 Specificity : 0.8667 (86.7%)
 AUC : 0.9222

Logistic Regression Coefficients (sorted by importance):

	Feature	Coefficient	Odds Ratio
Number of vessels	fluro_2	1.895464	6.655633
Number of vessels	fluro_1	1.509720	4.525466
	Chest pain type_4	1.152518	3.166155
	Sex	1.130622	3.097583
Number of vessels	fluro_3	0.990758	2.693276
	Thallium_7	0.890630	2.436665
	Max HR	-0.655009	0.519437
	ST depression	0.586686	1.798020
	Slope of ST_2	0.441198	1.554569
	Exercise angina	0.417169	1.517659
	EKG results_2	0.302807	1.353653
	Chest pain type_3	-0.299088	0.741494
	Cholesterol	0.274102	1.315349
	Chest pain type_2	0.205447	1.228074
	Slope of ST_3	0.160062	1.173583
	Age	-0.147987	0.862442
	EKG results_1	0.089105	1.093195
	Thallium_6	-0.074668	0.928051
	BP	0.009631	1.009678
	FBS over 120	-0.004310	0.995699

Accuracy: 84.0%. Sensitivity: 80.6%, Specificity: 86.7%. AUC = 0.922. The most important predictors are number of vessels, chest pain type 4, sex, and maximum heart rate.

Linear Discriminant Analysis (LDA)

```
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LDA
=====
```

Confusion Matrix:

	Predicted Absence	Predicted Presence
Actual Absence	38	7
Actual Presence	9	27

Accuracy : 0.8025 (80.2%)
Sensitivity : 0.7500 (75.0%)
Specificity : 0.8444 (84.4%)
AUC : 0.9111

Accuracy: 80.2%. Sensitivity: 75.0%, Specificity: 84.4%. AUC = 0.911. Slightly weaker than logistic regression.

Quadratic Discriminant Analysis (QDA)

```
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QDA
=====
```

Confusion Matrix:

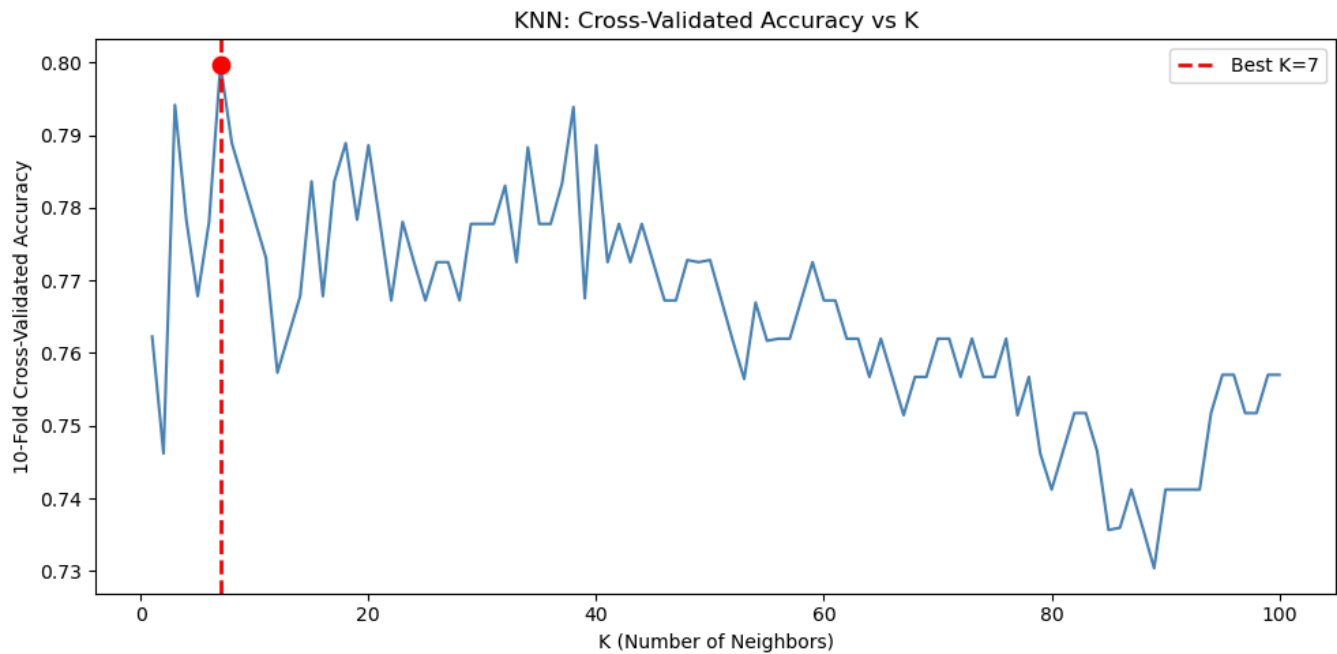
	Predicted Absence	Predicted Presence
Actual Absence	34	11
Actual Presence	10	26

Accuracy : 0.7407 (74.1%)
Sensitivity : 0.7222 (72.2%)
Specificity : 0.7556 (75.6%)
AUC : 0.8105

Accuracy: 74.1%. Sensitivity: 72.2%, Specificity: 75.6%. AUC = 0.811. Weakest model — the added flexibility of QDA introduces overfitting on this small dataset.

K-Nearest Neighbors (KNN)

KNN: 10-Fold Cross-Validation to Select Optimal K (1 to 100)...
Best K : 7 (10-fold CV Accuracy = 0.7997)



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KNN (K=7)

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Confusion Matrix:

	Predicted Absence	Predicted Presence
Actual Absence	38	7
Actual Presence	11	25

Accuracy : 0.7778 (77.8%)

Sensitivity : 0.6944 (69.4%)

Specificity : 0.8444 (84.4%)

AUC : 0.8491

Best K = 7. Accuracy: 77.8%, Sensitivity: 69.4%, Specificity: 84.4%. AUC = 0.849.

Naive Bayes

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Naive Bayes

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Confusion Matrix:

	Predicted Absence	Predicted Presence
Actual Absence	36	9
Actual Presence	7	29

Accuracy : 0.8025 (80.2%)

Sensitivity : 0.8056 (80.6%)

Specificity : 0.8000 (80.0%)

AUC : 0.8914

Accuracy: 80.2%, Sensitivity: 80.6%, Specificity: 80.0%. AUC = 0.891. Strong sensitivity, good option when the priority is catching as many disease cases as possible.

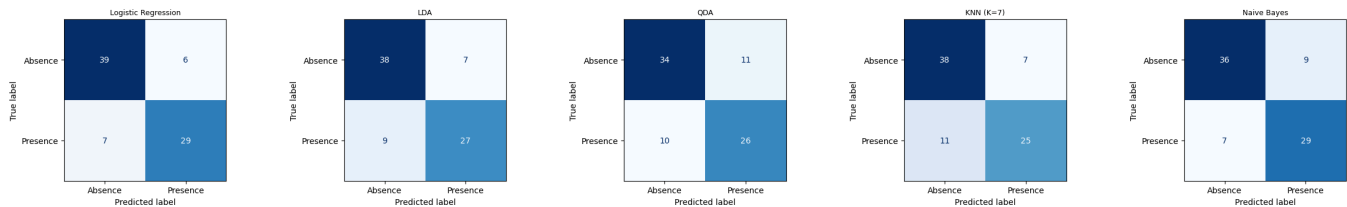
8. Final Model Comparison

We compare all five models across accuracy, sensitivity, specificity, and AUC using a summary table, confusion matrices, ROC curves, and bar charts.

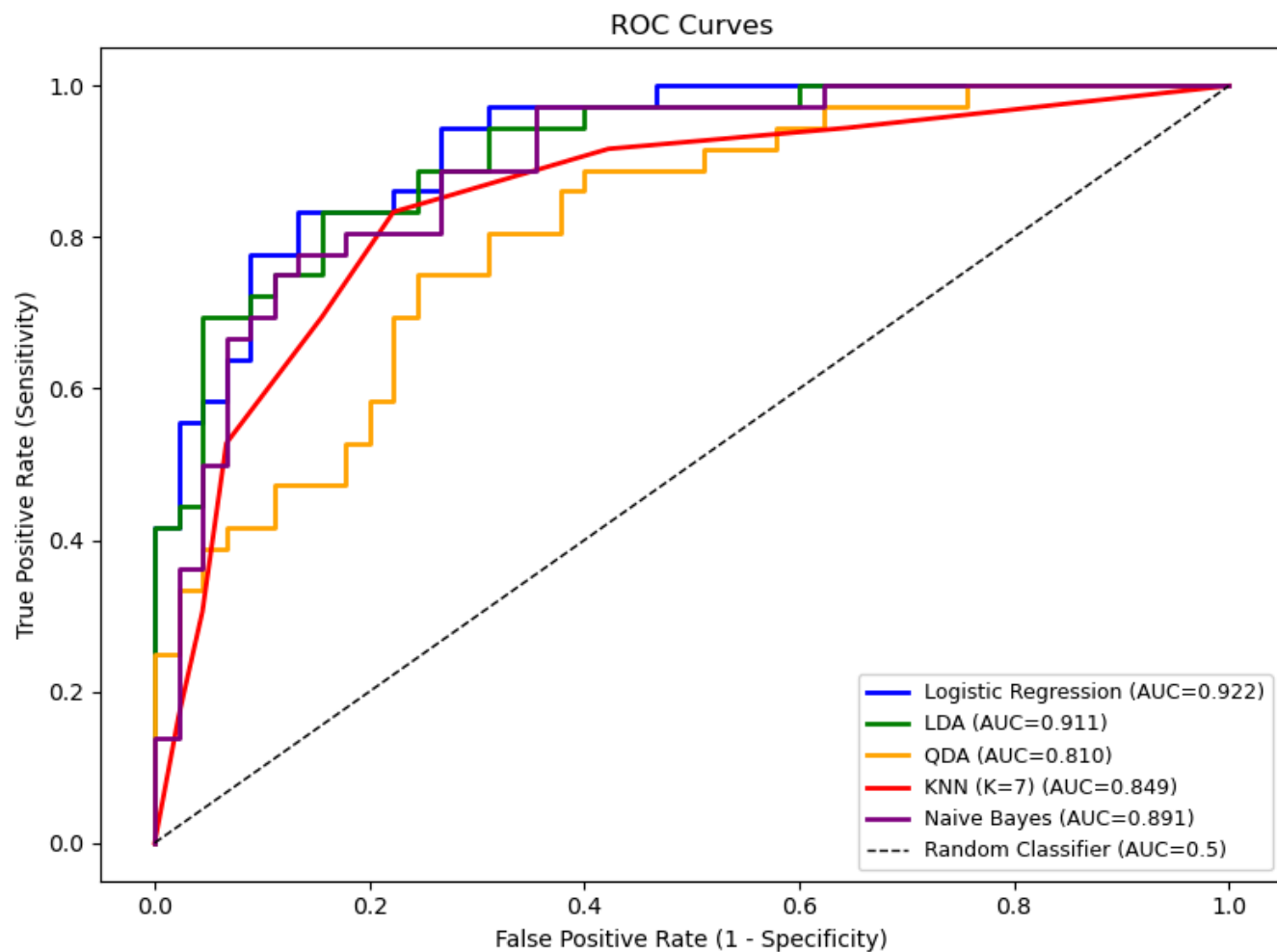
Final Model Comparison:

	Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression		0.8395	0.8056	0.8667	0.9222
	LDA	0.8025	0.7500	0.8444	0.9111
	QDA	0.7407	0.7222	0.7556	0.8105
	KNN (K=7)	0.7778	0.6944	0.8444	0.8491
	Naive Bayes	0.8025	0.8056	0.8000	0.8914

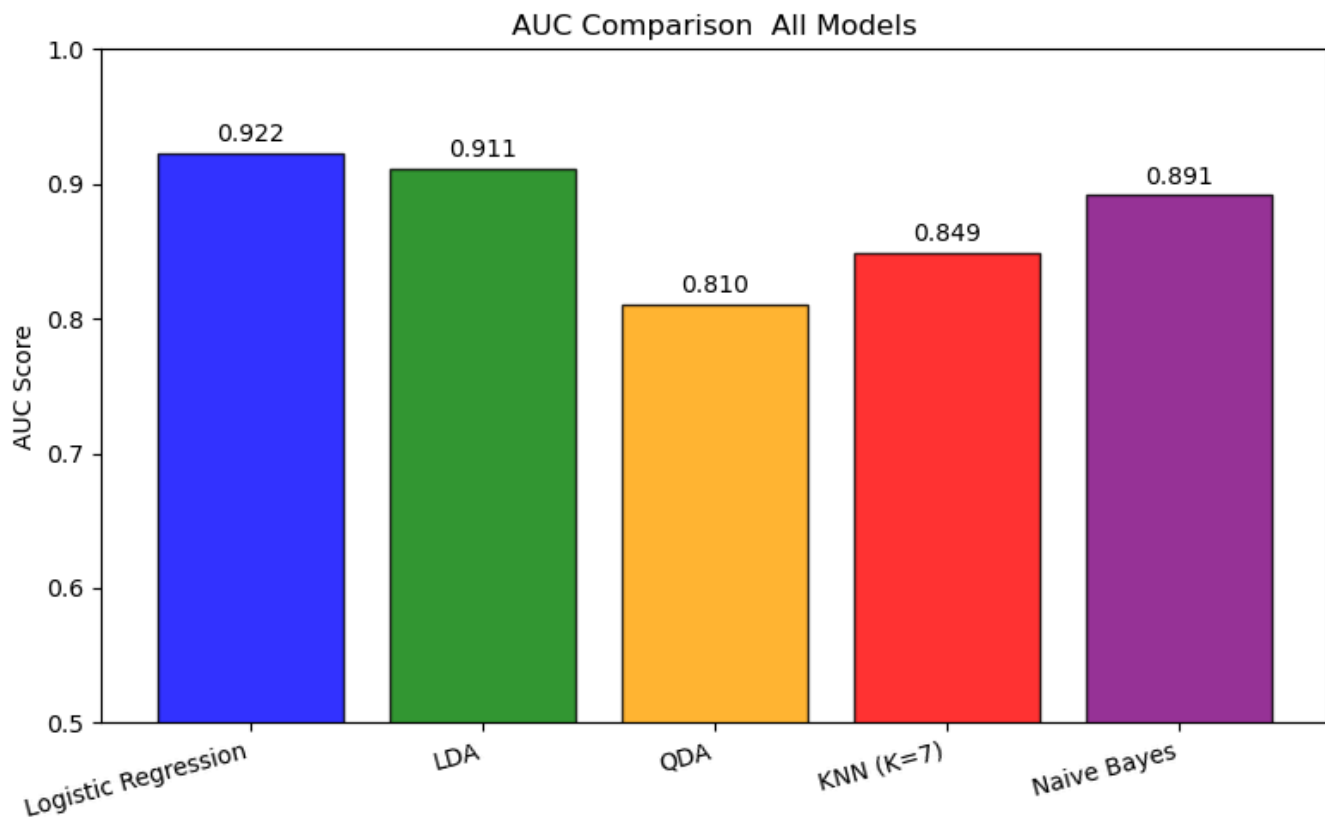
Confusion Matrices (Test Set)



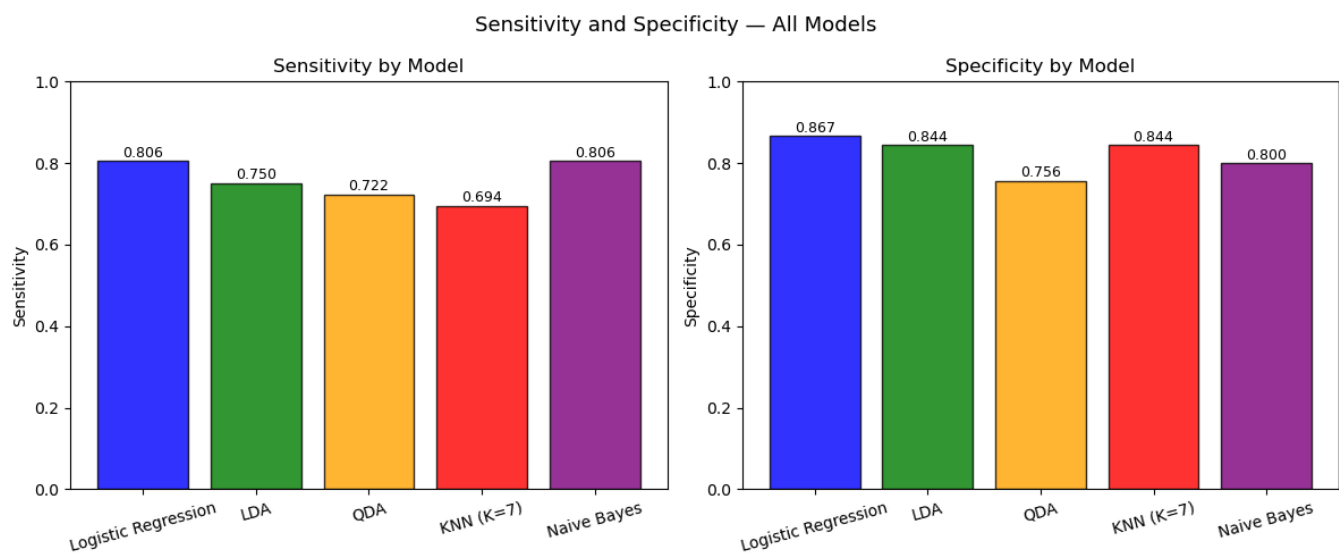
Logistic regression correctly identifies 29 out of 36 true heart disease cases and 39 out of 45 healthy patients, the best balance. KNN and QDA miss the most actual cases (11 and 10 respectively), making them the weakest options in a clinical setting where false negatives are costly.



The ROC curve shows the tradeoff between sensitivity and specificity at every possible classification threshold. A model closer to the top-left corner is better. The diagonal dashed line represents a random classifier with AUC = 0.5.



Logistic regression leads with AUC = 0.922, followed by LDA (0.911) and Naive Bayes (0.891). QDA scores the lowest at 0.811.



Done. Final summary:

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.8395	0.8056	0.8667	0.9222
LDA	0.8025	0.7500	0.8444	0.9111
QDA	0.7407	0.7222	0.7556	0.8105
KNN (K=7)	0.7778	0.6944	0.8444	0.8491
Naive Bayes	0.8025	0.8056	0.8000	0.8914

Logistic regression and Naive Bayes share the highest sensitivity (80.6%), they catch the most actual heart disease cases. KNN has the lowest sensitivity (69.4%), missing nearly 1 in 3 cases. For specificity, logistic regression leads at 86.7%, correctly ruling out the most healthy patients.

Conclusion

We applied and compared five classification models to predict heart disease presence from clinical measurements.

Logistic regression achieved the best overall performance with accuracy of 84.0% and AUC of 0.922. LDA and Naive Bayes followed closely. QDA performed worst, likely due to overfitting on the small sample of 270 patients. KNN was competitive but had the lowest sensitivity, making it the least suitable choice for a clinical problem where missing a true case matters most.

In terms of sensitivity, the most clinically relevant metric, logistic regression and Naive Bayes both correctly identified 80.6% of actual heart disease cases. The most significant predictors across models were number of fluoroscopy vessels, chest pain type, sex, and maximum heart rate.

Best model: Logistic Regression Accuracy: 84.0% | Sensitivity: 80.6% | Specificity: 86.7% | AUC: 0.922

Regression Analysis : CASP Dataset

Dataset: Physicochemical Properties of Protein Tertiary Structure

Target variable: F3 : Fractional area of exposed non-polar residue

Research question: Can we predict the hydrophobic surface exposure of a protein from its other physicochemical properties?

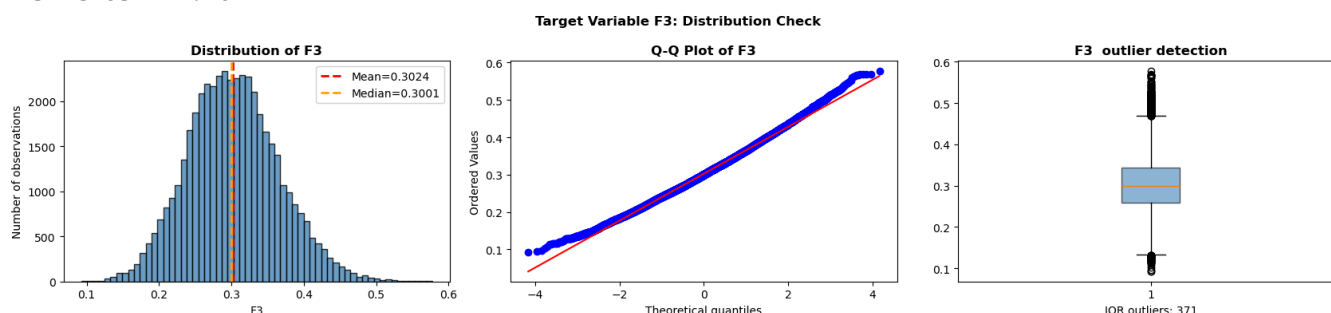
We begin by importing all required libraries and loading the dataset. The dataset contains 45,730 protein observations with 10 physicochemical variables. We verify there are no missing values before proceeding.

The dataset contains 45,730 proteins and 10 variables with no missing values. F3 ranges between 0.09 and 0.58, confirming it is a fraction bounded between 0 and 1. No data cleaning is required.

Checking the Distribution of F3

Before building any model, we need to check whether F3 follows a normal distribution. This matters because OLS regression assumes the errors will be normally distributed, and if the target variable itself is heavily skewed, the errors will be too. We use three checks: a histogram to see the shape, a Q-Q plot to compare against a perfect normal distribution, and the skewness value to measure asymmetry numerically.

F3 mean : 0.3024
F3 median : 0.3001
F3 skewness : 0.2379
F3 min : 0.0925
F3 max : 0.5777
F3 zeros : 0

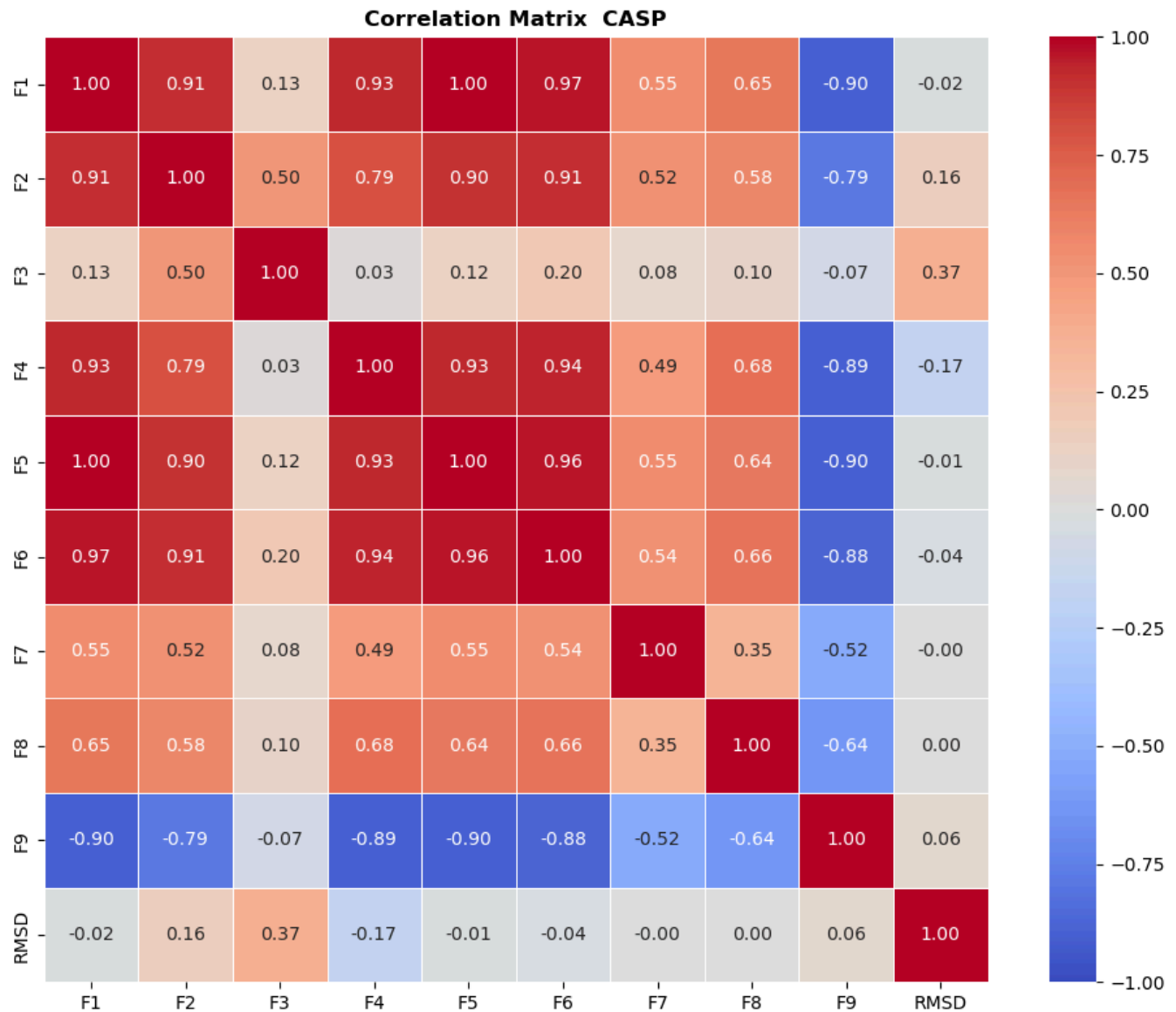


The mean (0.3024) and median (0.3001) are virtually identical, confirming symmetry with no extreme values pulling the average. The skewness of 0.24 is well within the acceptable range of ± 0.5 . The histogram shows a clean bell-shaped distribution and the Q-Q plot confirms points follow the diagonal closely. The boxplot identifies 371 IQR outliers, only 0.8% of 45,730 proteins, consistent with what a normal distribution naturally produces. No log-transformation is required. We work directly with F3.

2. Correlation Analysis

We examine correlations between all variables to identify multicollinearity before variable selection.

F3 skewness = 0.2379



F1, F2, F4, F5, F6 are all highly correlated ($r > 0.90$), they all measure protein size.

F1 and F5 are perfectly correlated ($r = 1.00$) and are removed as redundant.

RMSD is independent of all size variables ($r \approx 0$).

Final candidate predictors: **F2, F4, F6, F7, F8, F9, RMSD**.

3. Train / Test Split (70% / 30%)

We split the data to evaluate model generalisation on unseen proteins.

Training set : 32,011 proteins (70%)
 Test set : 13,719 proteins (30%)

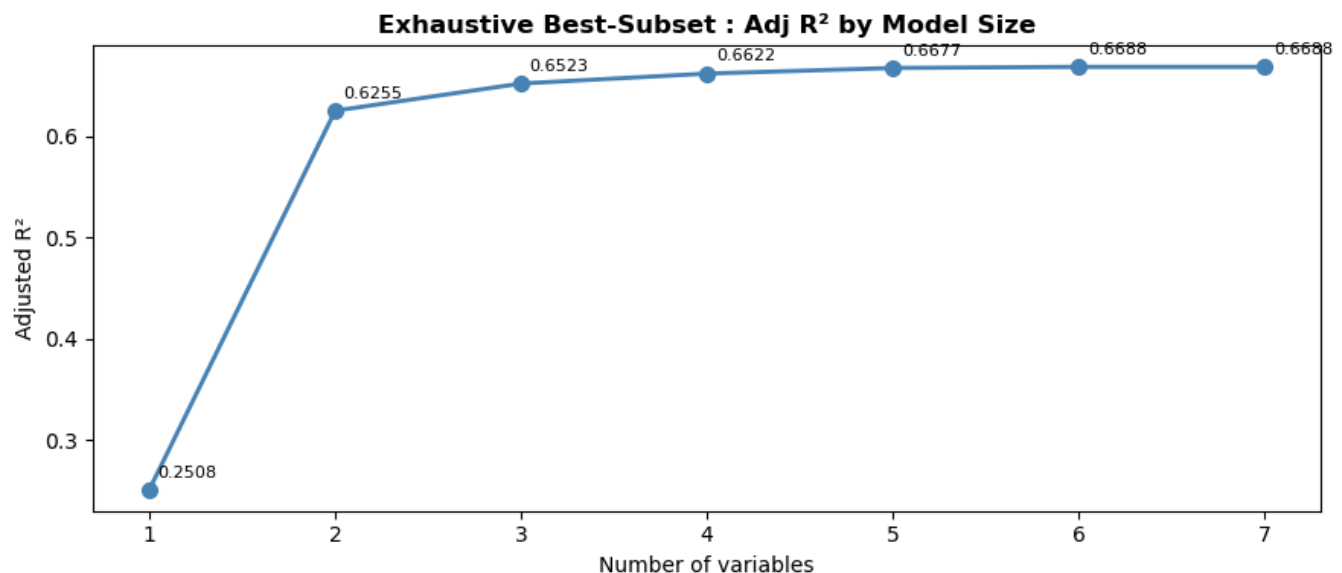
4. Exhaustive Best-Subset Selection

We test all 127 possible combinations of the 7 candidate variables and select the best model at each size using Adjusted R^2 .

```
1 var(s): Adj R²=0.25079 ['F2']
2 var(s): Adj R²=0.62549 ['F2', 'F6']
3 var(s): Adj R²=0.65233 ['F2', 'F6', 'F9']
4 var(s): Adj R²=0.66216 ['F2', 'F4', 'F6', 'F9']
5 var(s): Adj R²=0.66770 ['F2', 'F4', 'F6', 'F7', 'F9']
6 var(s): Adj R²=0.66881 ['F2', 'F4', 'F6', 'F7', 'F8', 'F9']
7 var(s): Adj R²=0.66880 ['F2', 'F4', 'F6', 'F7', 'F8', 'F9', 'RMSD']
```

→ Best subset: ['F2', 'F4', 'F6', 'F7', 'F8', 'F9']
 Adj R^2 : 0.66881

Adjusted R^2 improves up to 6 variables then plateaus. Adding RMSD as a 7th variable provides no improvement (Adj R^2 drops from 0.66881 to 0.66880). Optimal model: **F2, F4, F6, F7, F8, F9**.



The elbow curve confirms the optimal model size is 6 variables.

5. Chosen Model

We fit the 6-variable OLS regression selected by exhaustive search.

OLS Regression Results

Dep. Variable:		F3		R-squared:		0.669
Model:		OLS		Adj. R-squared:		0.669
Method:		Least Squares		F-statistic:		1.077e+04
Date:		Thu, 19 Mar 2026		Prob (F-statistic):		0.00
Time:		20:17:00		Log-Likelihood:		60787.
No. Observations:		32011		AIC:		-1.216e+05
Df Residuals:		32004		BIC:		-1.215e+05
Df Model:		6				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.1471	0.004	40.377	0.000	0.140	0.154
F2	7.296e-05	3.62e-07	201.564	0.000	7.22e-05	7.37e-05
F4	-0.0004	1.28e-05	-34.042	0.000	-0.000	-0.000
F6	-0.0007	1.35e-05	-50.710	0.000	-0.001	-0.001
F7	-2.688e-06	1.17e-07	-23.015	0.000	-2.92e-06	-2.46e-06
F8	5.114e-05	4.91e-06	10.408	0.000	4.15e-05	6.08e-05
F9	0.0025	7.9e-05	31.738	0.000	0.002	0.003
Omnibus:		4275.461	Durbin-Watson:		1.999	
Prob(Omnibus):		0.000	Jarque-Bera (JB):		33629.793	
Skew:		-0.399	Prob(JB):		0.00	
Kurtosis:		7.958	Cond. No.		9.85e+04	

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 9.85e+04. This might indicate that there are strong multicollinearity or other numerical problems.

The model explains 66.9% of the variation in F3.

All six predictors are highly significant ($p \approx 0$). The condition number warning reflects known multicollinearity among the size variables acknowledged as a limitation but does not invalidate the model at $n = 32,011$.

6. Scaled Model : Predictor Importance

We standardise all variables to compare predictor importance on a common scale.

OLS Regression Results

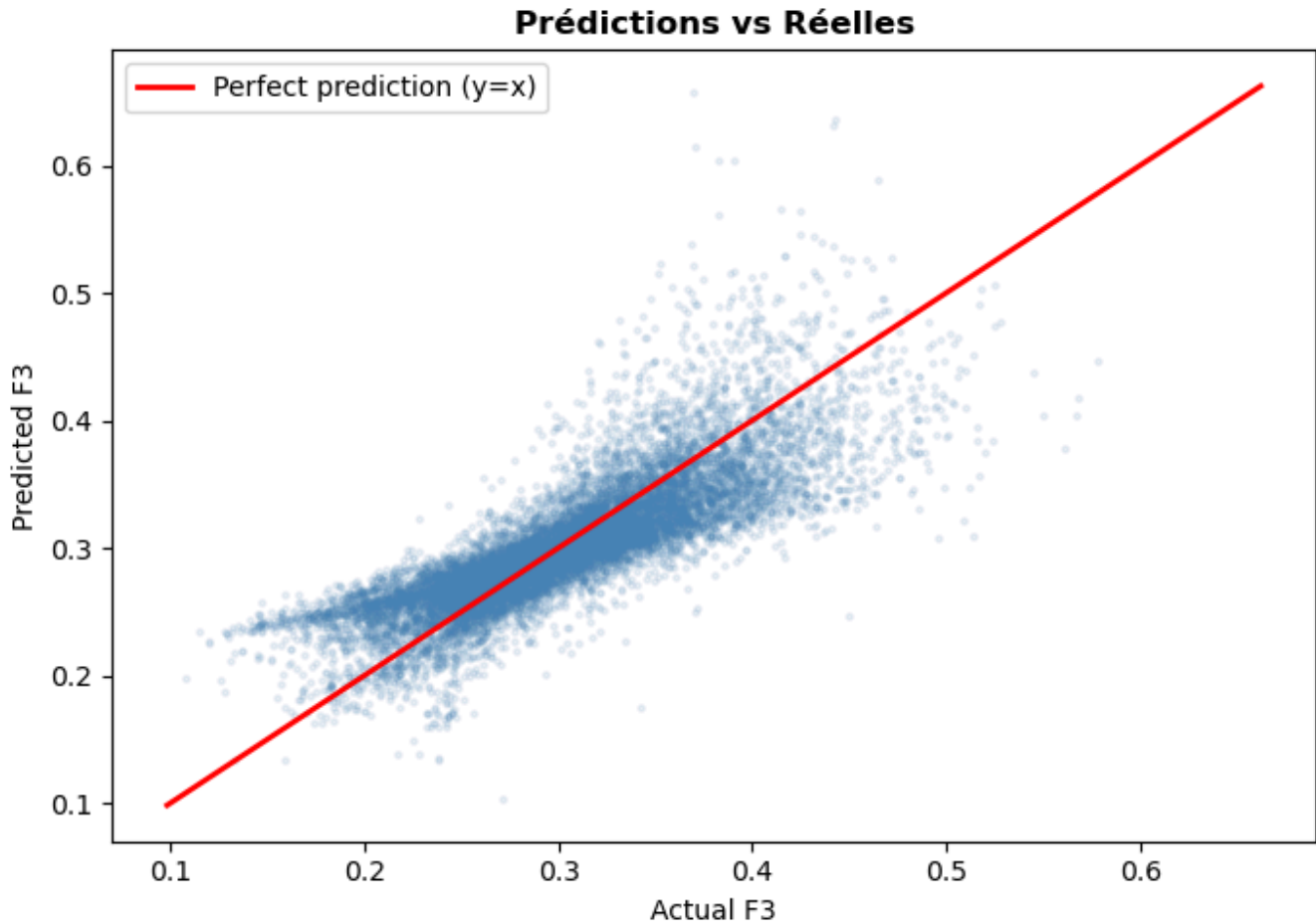
Dep. Variable:	F3	R-squared:	0.669
Model:	OLS	Adj. R-squared:	0.669
Method:	Least Squares	F-statistic:	1.077e+04
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	0.00
Time:	20:17:29	Log-Likelihood:	-27731.
No. Observations:	32011	AIC:	5.548e+04
Df Residuals:	32004	BIC:	5.553e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	8.544e-17	0.003	2.66e-14	1.000	-0.006	0.006
F2	1.7023	0.008	201.564	0.000	1.686	1.719
F4	-0.3841	0.011	-34.042	0.000	-0.406	-0.362
F6	-0.7612	0.015	-50.710	0.000	-0.791	-0.732
F7	-0.0877	0.004	-23.015	0.000	-0.095	-0.080
F8	0.0459	0.004	10.408	0.000	0.037	0.055
F9	0.2388	0.008	31.738	0.000	0.224	0.254

Omnibus:	4275.461	Durbin-Watson:	1.999
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33629.793
Skew:	-0.399	Prob(JB):	0.00
Kurtosis:	7.958	Cond. No.	12.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



F2 is the dominant predictor ($|\beta| = 1.703$), more than double the next variable F6 (0.768). Ranking: F2 > F6 > F4 > F9 > F7 > F8. R^2 remains 0.669, scaling only changes coefficient interpretation, not model performance.

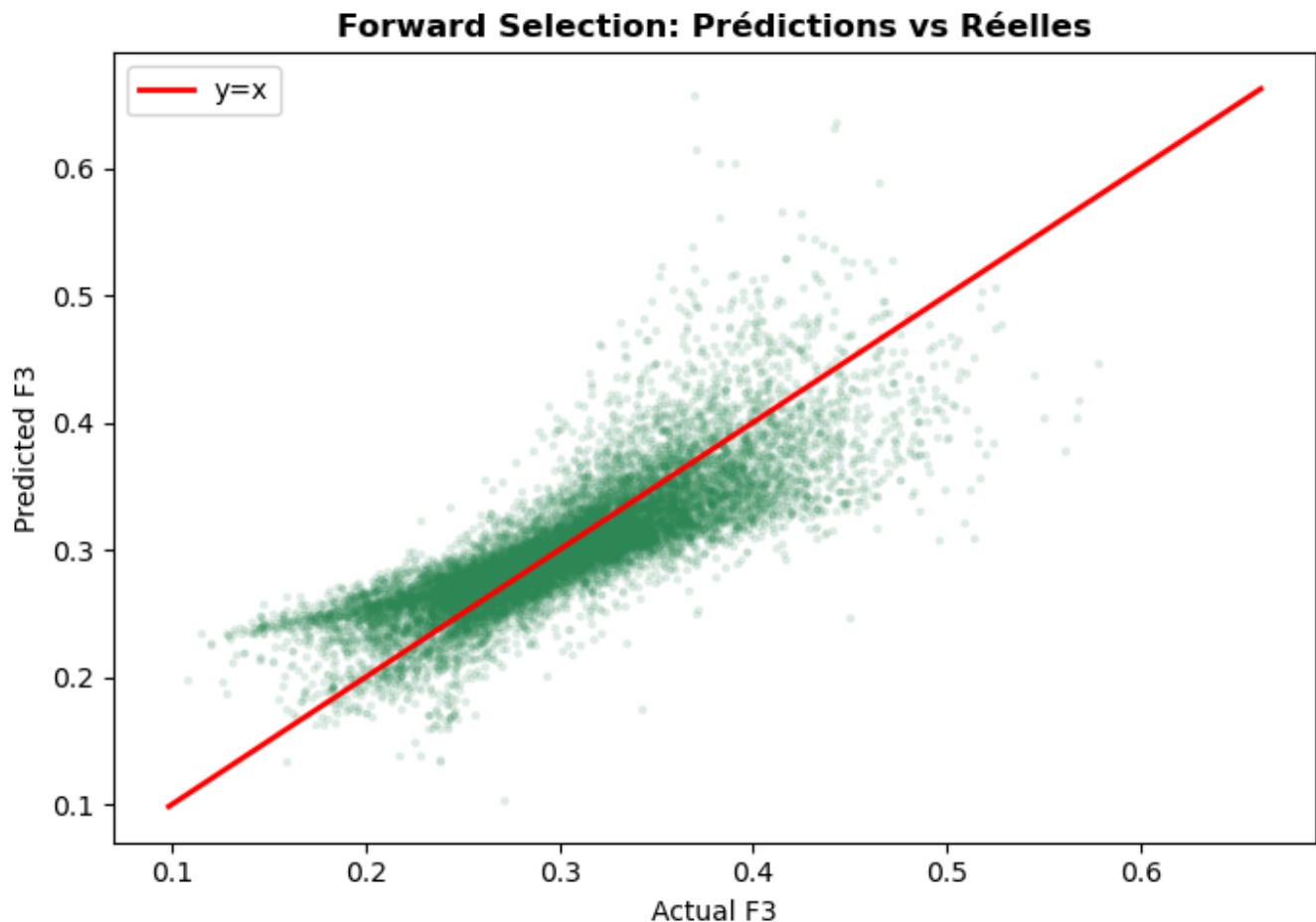
7. Forward Selection

Starting from an empty model, we add variables one at a time by lowest p-value, stopping when nothing remaining is significant ($p < 0.05$).

Forward selection steps:

+ F2	p=0.0000e+00	Adj R ² =0.25079
+ F4	p=0.0000e+00	Adj R ² =0.61514
+ F6	p=0.0000e+00	Adj R ² =0.64963
+ F9	p=1.0960e-255	Adj R ² =0.66216
+ F7	p=2.9425e-117	Adj R ² =0.66770
+ F8	p=2.5071e-25	Adj R ² =0.66881
STOP no remaining variable has $p < 0.05$		

Forward selected vars: ['F2', 'F4', 'F6', 'F9', 'F7', 'F8']



Six variables added in order: F2, F4, F6, F9, F7, F8. RMSD was never added, its p-value exceeded 0.05 once the other six were present. Result identical to exhaustive search.

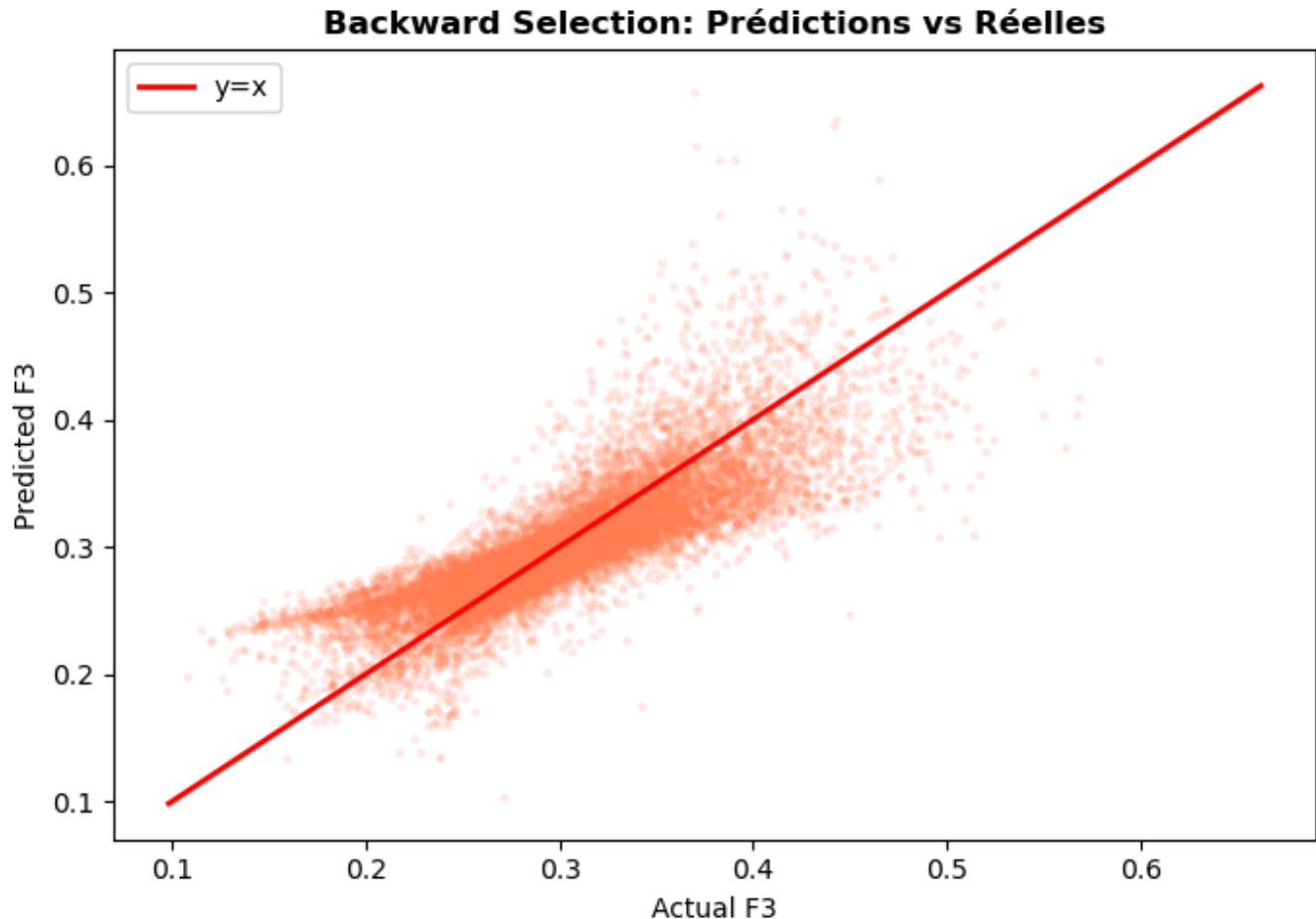
8. Backward Selection

Starting from all 7 variables, we remove the least significant one at each step until all remaining have $p < 0.05$.

Backward selection steps:

- RMSD $p=0.7203$ Adj $R^2=0.66881$
STOP all remaining variables have $p < 0.05$

Backward selected vars: ['F2', 'F4', 'F6', 'F7', 'F8', 'F9']



Only RMSD was removed ($p = 0.72$). All other variables were retained. Final result: F2, F4, F6, F7, F8, F9, identical to exhaustive and forward selection. All three methods converging on the same variables confirms model stability.

9. Baseline Performance : Test Set

We evaluate all three models on the 13,719 held-out test proteins using RMSE and R^2 .

Model	RMSE	R^2
Exhaustive	0.035394	0.681393
Forward	0.035394	0.681393
Backward	0.035394	0.681393

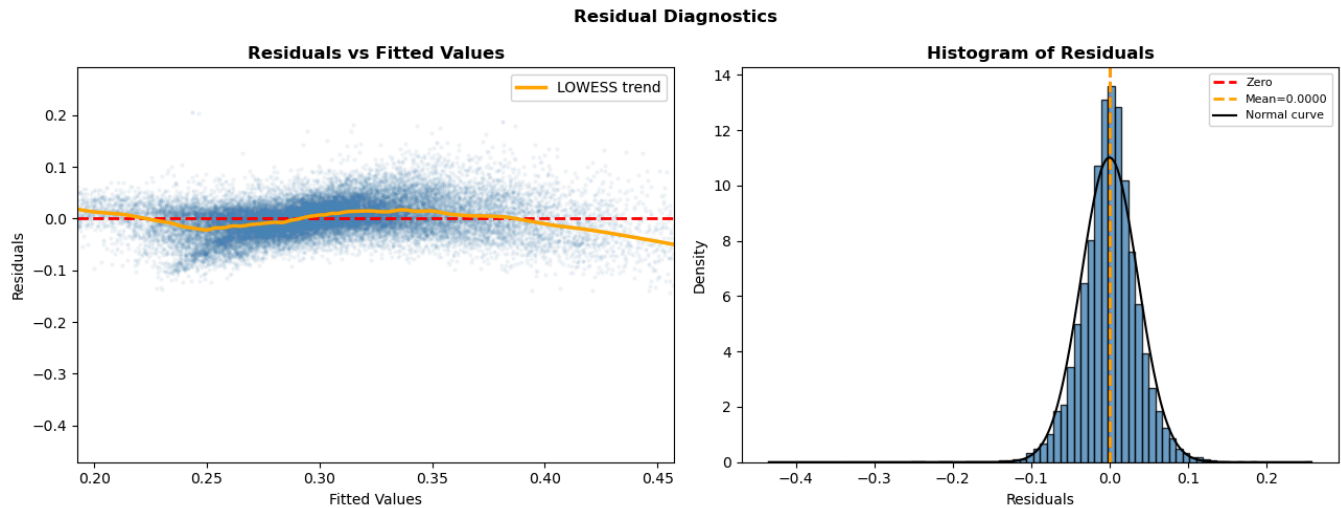
All three models produce identical results ($R^2 = 0.681$, $RMSE = 0.035$) since they selected the same variables.

Test R^2 slightly exceeds training R^2 (0.669), confirming no overfitting. Average prediction error of 0.035 F3 units $\approx 7\%$ of the total F3 range.

10. Residual Diagnostics

We verify the two key OLS assumptions: homoscedasticity and normality of residuals.

Mean of residuals : 5.68e-16
Median of residuals : 0.000644



Mean of residuals ≈ 0 (OLS guarantee). Histogram is approximately bell-shaped and centred at zero, normality satisfied. Kurtosis = 7.958 indicates a slightly peaked distribution with heavier tails than normal. The LOWESS trend shows a gentle downward slope across fitted values, signalling mild non-linearity in F2. This motivates the polynomial model in the next step.

11. Polynomial Regression : F2²

The residual plot revealed non-linearity attributable to F2, the dominant predictor. At large protein surfaces, fractional hydrophobic exposure saturates rather than increasing linearly. We add F2² to capture this. T

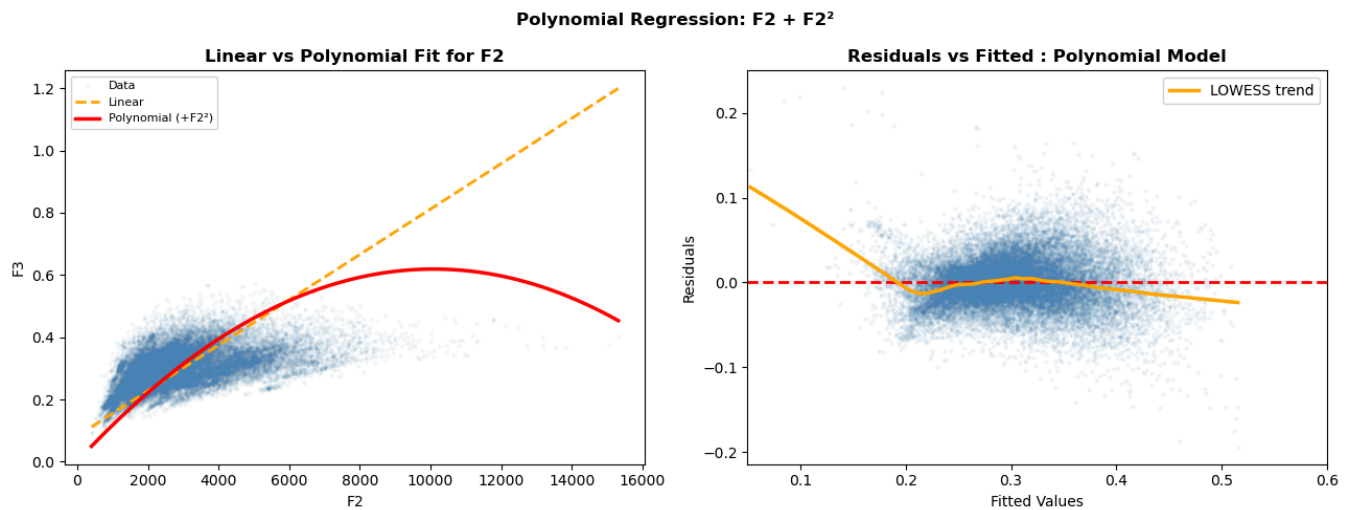
OLS Regression Results

Dep. Variable:		F3	R-squared:		0.774	
Model:		OLS	Adj. R-squared:		0.774	
Method:		Least Squares		F-statistic:		1.566e+04
Date:		Thu, 19 Mar 2026		Prob (F-statistic):		0.00
Time:		20:21:06		Log-Likelihood:		66899.
No. Observations:		32011		AIC:		-1.338e+05
Df Residuals:		32003		BIC:		-1.337e+05
Df Model:		7				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.0364	0.003	-10.815	0.000	-0.043	-0.030
F2	0.0001	5.04e-07	242.950	0.000	0.000	0.000
F4	-0.0004	1.05e-05	-41.140	0.000	-0.000	-0.000
F6	-0.0006	1.12e-05	-49.961	0.000	-0.001	-0.001
F7	-2.368e-06	9.65e-08	-24.527	0.000	-2.56e-06	-2.18e-06
F8	5.131e-05	4.06e-06	12.639	0.000	4.34e-05	5.93e-05
F9	0.0049	6.82e-05	72.021	0.000	0.005	0.005
F2_sq	-6.069e-09	4.97e-11	-122.001	0.000	-6.17e-09	-5.97e-09
Omnibus:		2154.694	Durbin-Watson:		2.003	
Prob(Omnibus):		0.000	Jarque-Bera (JB):		6954.122	
Skew:		0.318	Prob(JB):		0.00	
Kurtosis:		5.193	Cond. No.		3.31e+08	

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.31e+08. This might indicate that there are strong multicollinearity or other numerical problems.



$F2^2$ is highly significant ($p \approx 0$, $\beta = -6.07e-09$). The negative coefficient confirms a concave relationship, exposure increases then saturates. Adjusted R^2 improves from 0.669 to 0.774.

The linear model was predicting $F3 > 1.0$ at high $F2$ values, biologically impossible. The polynomial curve stays within realistic bounds and the residual LOWESS is considerably flatter.

12. Interaction Model: $RMSD \times F2$

We test whether the effect of $RMSD$ on $F3$ depends on $F2$, whether the two variables act synergistically. Using $RMSD * F2$ automatically includes both main effects, satisfying the Hierarchy Principle.

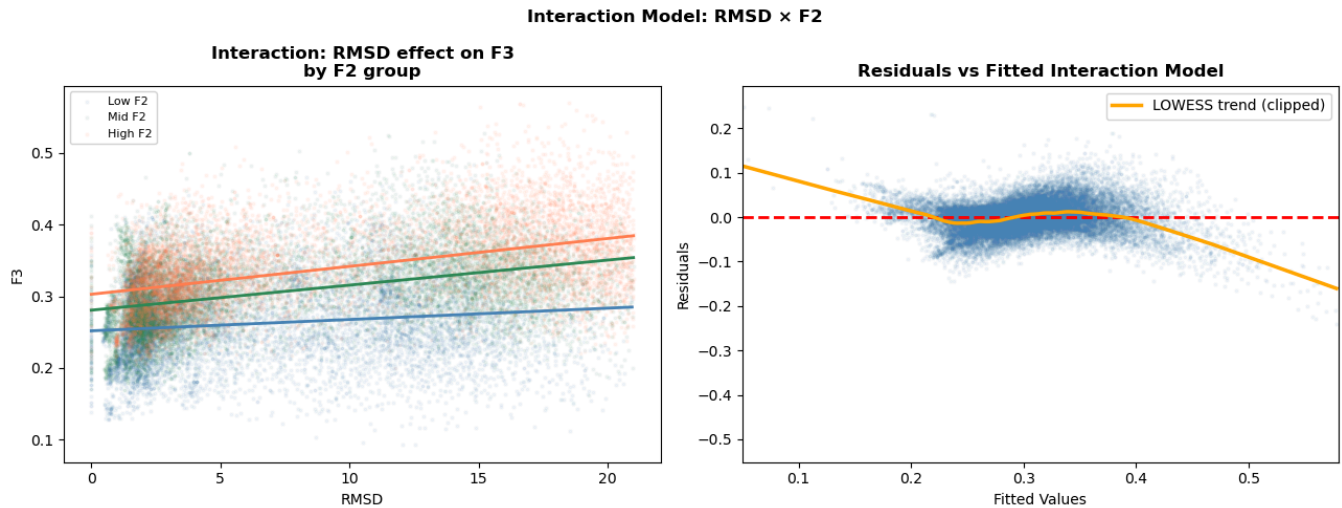
OLS Regression Results

Dep. Variable:		F3		R-squared:		0.683
Model:		OLS		Adj. R-squared:		0.683
Method:		Least Squares		F-statistic:		8613.
Date:		Thu, 19 Mar 2026		Prob (F-statistic):		0.00
Time:		20:19:49		Log-Likelihood:		61477.
No. Observations:		32011		AIC:		-1.229e+05
Df Residuals:		32002		BIC:		-1.229e+05
Df Model:		8				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.1373	0.004	38.011	0.000	0.130	0.144
RMSD	0.0025	7.71e-05	32.470	0.000	0.002	0.003
F2	8.204e-05	4.47e-07	183.410	0.000	8.12e-05	8.29e-05
RMSD:F2	-8.244e-07	2.2e-08	-37.550	0.000	-8.67e-07	-7.81e-07
F4	-0.0005	1.33e-05	-39.982	0.000	-0.001	-0.001
F6	-0.0007	1.32e-05	-51.353	0.000	-0.001	-0.001
F7	-2.686e-06	1.14e-07	-23.482	0.000	-2.91e-06	-2.46e-06
F8	5.065e-05	4.84e-06	10.456	0.000	4.12e-05	6.01e-05
F9	0.0023	7.77e-05	29.475	0.000	0.002	0.002
Omnibus:		5674.259	Durbin-Watson:		2.001	
Prob(Omnibus):		0.000	Jarque-Bera (JB):		70353.012	
Skew:		-0.483	Prob(JB):		0.00	
Kurtosis:		10.198	Cond. No.		6.70e+05	

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 6.7e+05. This might indicate that there are strong multicollinearity or other numerical problems.



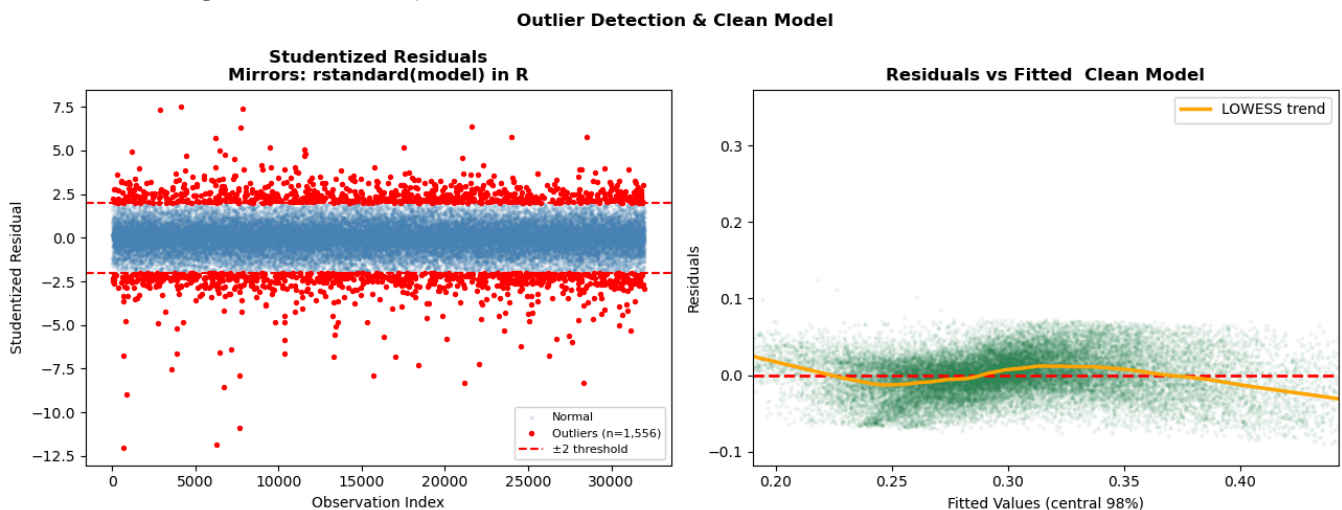
The interaction term RMSD:F2 is significant ($p \approx 0$, $\beta = -8.24e-07$). The negative sign indicates dampening, RMSD's effect on F3 weakens as F2 increases. The three F2 groups show different slopes in the left panel, confirming the interaction visually. Adj R^2 gain is significant but modest compared to the polynomial model.

13. Outlier Regression

We compute studentized residuals and flag proteins with $|e^*| > 2$ as outliers. We refit the model without them and compare performance.

```

Training proteins   : 32,011
Flagged ( $|e^*| > 2$ ) : 1,556 (4.9%)
Clean training set  : 30,455 proteins
  
```



1,556 outliers flagged (4.9%), consistent with the expected ~5% from a normal distribution. Removing them improves training R^2 (0.669 to 0.758) but reduces test R^2 (0.681 to 0.674). The outliers are genuine biological variation, likely intrinsically disordered or membrane proteins not errors. The original model is preferred for real-world prediction.

14. Final Model Comparison

Backward model — Test R^2 : 0.681393 RMSE: 0.035394

Clean model — Test R^2 : 0.67405 RMSE: 0.035799

Train R^2 original : 0.66887

Train R^2 clean : 0.75812

	Model	Train Adj R^2	Test R^2	Test RMSE
	Exhaustive (baseline)	0.66881	0.681393	0.035394
	Forward (baseline)	0.66881	0.681393	0.035394
	Backward (baseline)	0.66881	0.681393	0.035394
	Polynomial + $F2^2$ (Step 12)	0.77394	0.775288	0.029724
	Interaction $RMSD \times F2$ (Step 13)	0.68277	0.693736	0.034701
	Clean Backward (Step 14)	0.75807	0.674050	0.035799

The polynomial model ($F2^2$) is the best performer: Test R^2 = 0.775, RMSE = 0.030, a 9.4 pp improvement over baseline.

Training and test R^2 are nearly identical (0.774 vs 0.775), confirming no overfitting.

Best model: $F3 \sim F2 + F2^2 + F4 + F6 + F7 + F8 + F9$

The non-linear saturation effect of $F2$ on $F3$ was the key structural finding of this analysis.

Conclusion

Our analysis applied multiple linear regression to predict the fractional hydrophobic surface exposure ($F3$) of 45,730 proteins from their physicochemical properties.

Variable selection. All three methods, exhaustive search, forward selection, and backward elimination independently selected the same six variables: $F2$, $F4$, $F6$, $F7$, $F8$, $F9$. Out of 127 possible combinations, this convergence confirms that the signal in this dataset is strong and consistent across methods.

Non-linearity. The biggest improvement came from adding $F2^2$. Since $F3$ is a fraction bounded between 0 and 1, a straight line eventually predicts biologically impossible values above 1.0 for large proteins. The polynomial term corrected this and raised test R^2 from 0.681 to 0.775 a gain of nearly 10 percentage points.

Outliers. Removing statistical outliers improved training fit but reduced test performance. This is an important finding: in biology, outliers are not errors. They represent genuinely unusual proteins, intrinsically disordered proteins or membrane proteins that a real-world model must be able to handle. Deleting them produces an artificially clean model that fails on real data.

Limitation. The high R^2 of 0.775 partly reflects the nature of this dataset, the predictor variables and $F3$ are all derived from the same 3D atomic coordinates. They share a geometric origin, which inflates their correlations. The model is statistically valid but this structural relationship between variables should be acknowledged when interpreting predictive performance.

Final model: $F3 \sim F2 + F2^2 + F4 + F6 + F7 + F8 + F9$

Test $R^2 = 0.775$ | RMSE = 0.030

Regression Analysis : Heart Disease Dataset

Target variable: Max HR (Maximum heart rate achieved during exercise)

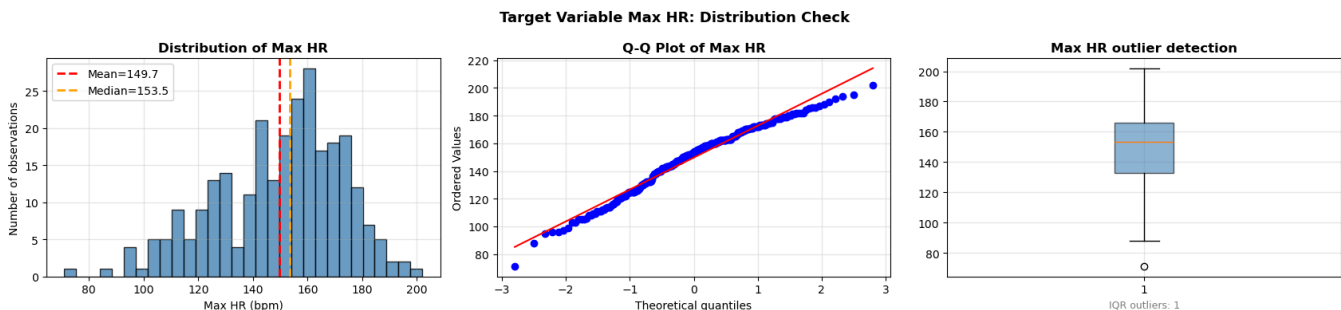
Question: Can we predict maximum heart rate from other cardiovascular and demographic measurements?

We load the dataset, rename columns for compatibility with Python formulas, and verify the structure before modelling.

270 patients, 14 variables, no missing values. Column names have been simplified to remove spaces for compatibility with statsmodels formulas.

1. Target Variable Distribution Check

We verify that Max HR satisfies the normality assumption required by OLS regression before any modelling.



The histogram shows a roughly bell-shaped distribution with a slight left skew. Mean (149.7) and median (153.5) are close but not identical. The Q-Q plot shows mild deviation from normality at the lower tail. The boxplot flags only 1 IQR outlier. Overall Max HR is approximately normal and no transformation is needed.

2. Log Transform Decision

```
Max HR skewness = -0.5248
Max HR mean     = 149.68
Max HR median   = 153.50
Max HR zeros    = 0
```

Skewness = -0.52, within the acceptable range of ± 0.5 . No zeros. No log-transformation needed. We work directly with Max HR.

3. Train / Test Split (70% / 30%)

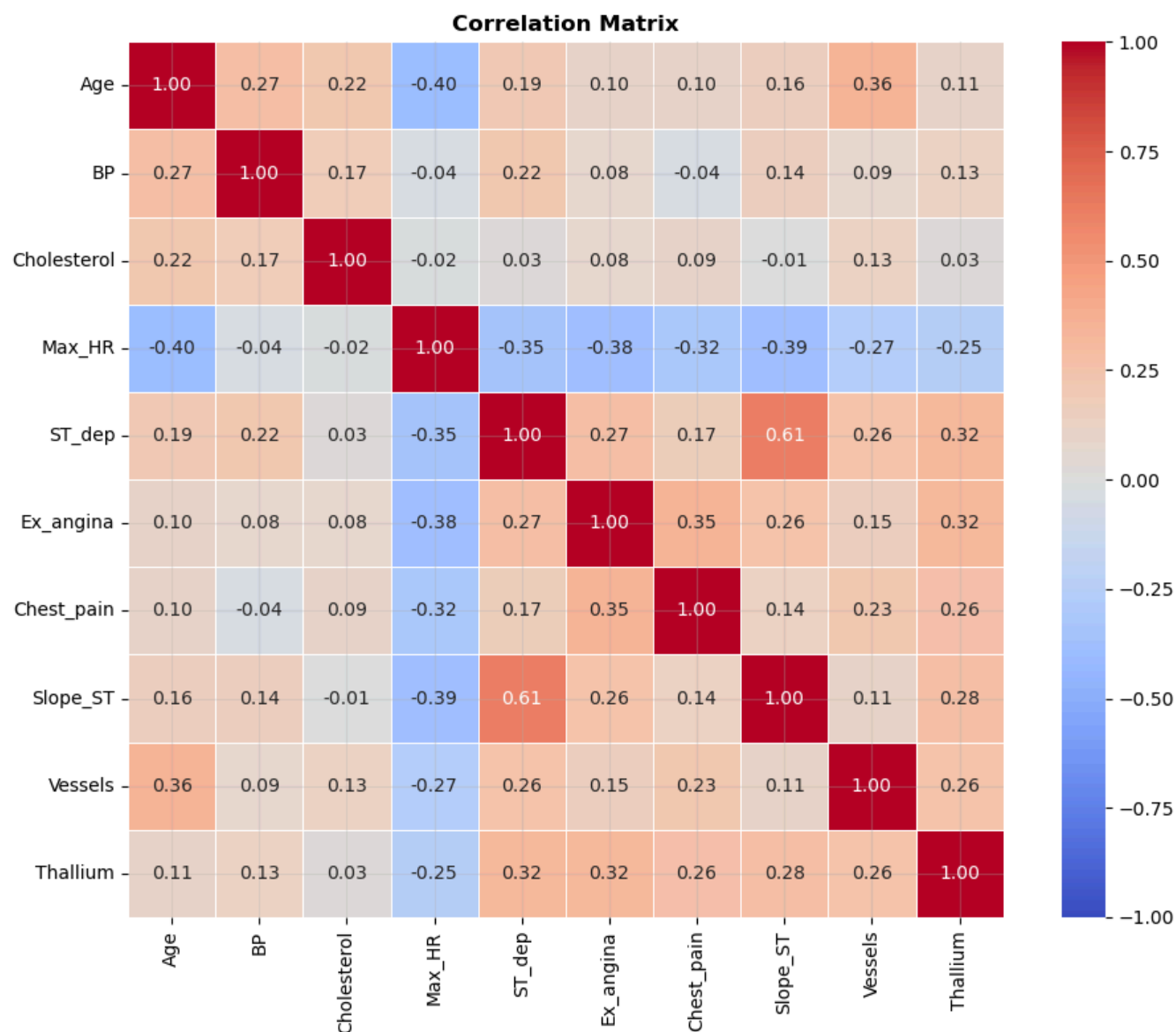
Training set : 189 patients

Test set : 81 patients

189 patients in training, 81 in the test set. All model fitting uses training data only.

4. Correlation Analysis

We examine correlations between all variables to identify which predictors are related to Max HR and detect any multicollinearity.



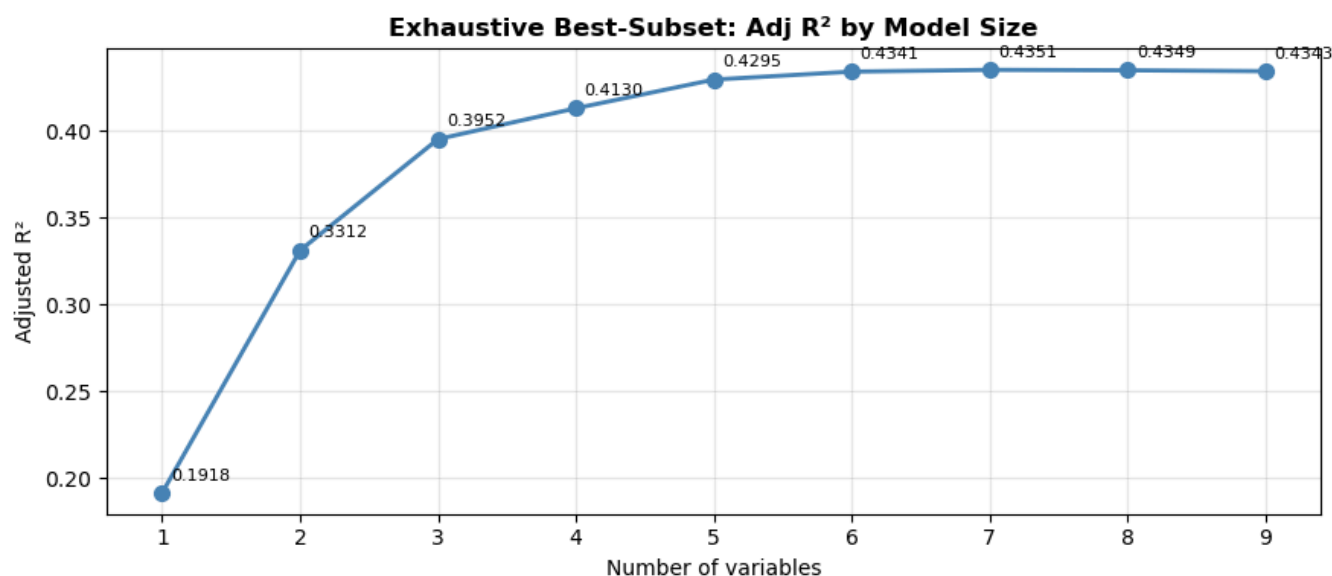
Age has the strongest correlation with Max HR ($r = -0.40$) — older patients achieve lower maximum heart rates. Exercise angina, slope of ST, and ST depression also show moderate negative correlations. BP and

Cholesterol are nearly uncorrelated with Max HR. No two predictors are perfectly collinear so all 9 candidates are kept for variable selection.

5. Exhaustive Best-Subset Selection

We test all 511 possible combinations of the 9 candidate variables ($2^9 - 1$) and identify the best model at each size using Adjusted R^2 .

```
" 1 var(s): Adj R²=0.19180 ['Ex_angina']"
" 2 var(s): Adj R²=0.33124 ['Age', 'Ex_angina']"
" 3 var(s): Adj R²=0.39524 ['Age', 'Ex_angina', 'Slope_ST']"
" 4 var(s): Adj R²=0.41300 ['Age', 'Ex_angina', 'Slope_ST', 'Vessels']"
" 5 var(s): Adj R²=0.42950 ['Age', 'BP', 'Ex_angina', 'Slope_ST', 'Vessels']"
" 6 var(s): Adj R²=0.43408 ['Age', 'BP', 'Ex_angina', 'Chest_pain', 'Slope_ST', 'Vessels']"
" 7 var(s): Adj R²=0.43510 ['Age', 'BP', 'ST_dep', 'Ex_angina', 'Chest_pain', 'Slope_ST',
'Vessels']"
" 8 var(s): Adj R²=0.43485 ['Age', 'BP', 'ST_dep', 'Ex_angina', 'Chest_pain', 'Slope_ST',
'Vessels', 'Thallium']"
" 9 var(s): Adj R²=0.43433 ['Age', 'BP', 'Cholesterol', 'ST_dep', 'Ex_angina', 'Chest_pai
n', 'Slope_ST', 'Vessels', 'Thallium']"
"Best subset: ['Age', 'BP', 'ST_dep', 'Ex_angina', 'Chest_pain', 'Slope_ST', 'Vessels'] Adj
R²=0.43510"
```



Adjusted R^2 improves up to 7 variables then plateaus. The elbow is at 7 variables. Best subset: Age, BP, ST_dep, Ex_angina, Chest_pain, Slope_ST, Vessels with Adj $R^2 = 0.435$. Unlike the CASP dataset, the gains are modest — the signal between these cardiac variables and Max HR is weak.

6. Chosen Model and Predictor Importance

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.456
Model:	OLS	Adj. R-squared:	0.435
Method:	Least Squares	F-statistic:	21.69
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	4.12e-21
Time:	20:09:17	Log-Likelihood:	-812.35
No. Observations:	189	AIC:	1641.
Df Residuals:	181	BIC:	1667.
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	198.9298	12.994	15.310	0.000	173.291	224.568
Age	-0.8605	0.166	-5.191	0.000	-1.188	-0.533
BP	0.2020	0.079	2.570	0.011	0.047	0.357
ST_dep	-1.7423	1.511	-1.153	0.250	-4.723	1.239
Ex_angina	-12.5615	3.274	-3.837	0.000	-19.022	-6.102
Chest_pain	-2.3957	1.565	-1.531	0.128	-5.484	0.693
Slope_ST	-9.2492	2.670	-3.464	0.001	-14.517	-3.981
Vessels	-3.1813	1.559	-2.040	0.043	-6.258	-0.105
Omnibus:	12.572	Durbin-Watson:	1.913			
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.272			
Skew:	-0.575	Prob(JB):	0.00131			
Kurtosis:	3.602	Cond. No.	1.42e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.42e+03. This might indicate that there are strong multicollinearity or other numerical problems.

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.456
Model:	OLS	Adj. R-squared:	0.435
Method:	Least Squares	F-statistic:	21.69
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	4.12e-21
Time:	20:09:17	Log-Likelihood:	-210.12
No. Observations:	189	AIC:	436.2
Df Residuals:	181	BIC:	462.2
Df Model:	7		
Covariance Type:	nonrobust		

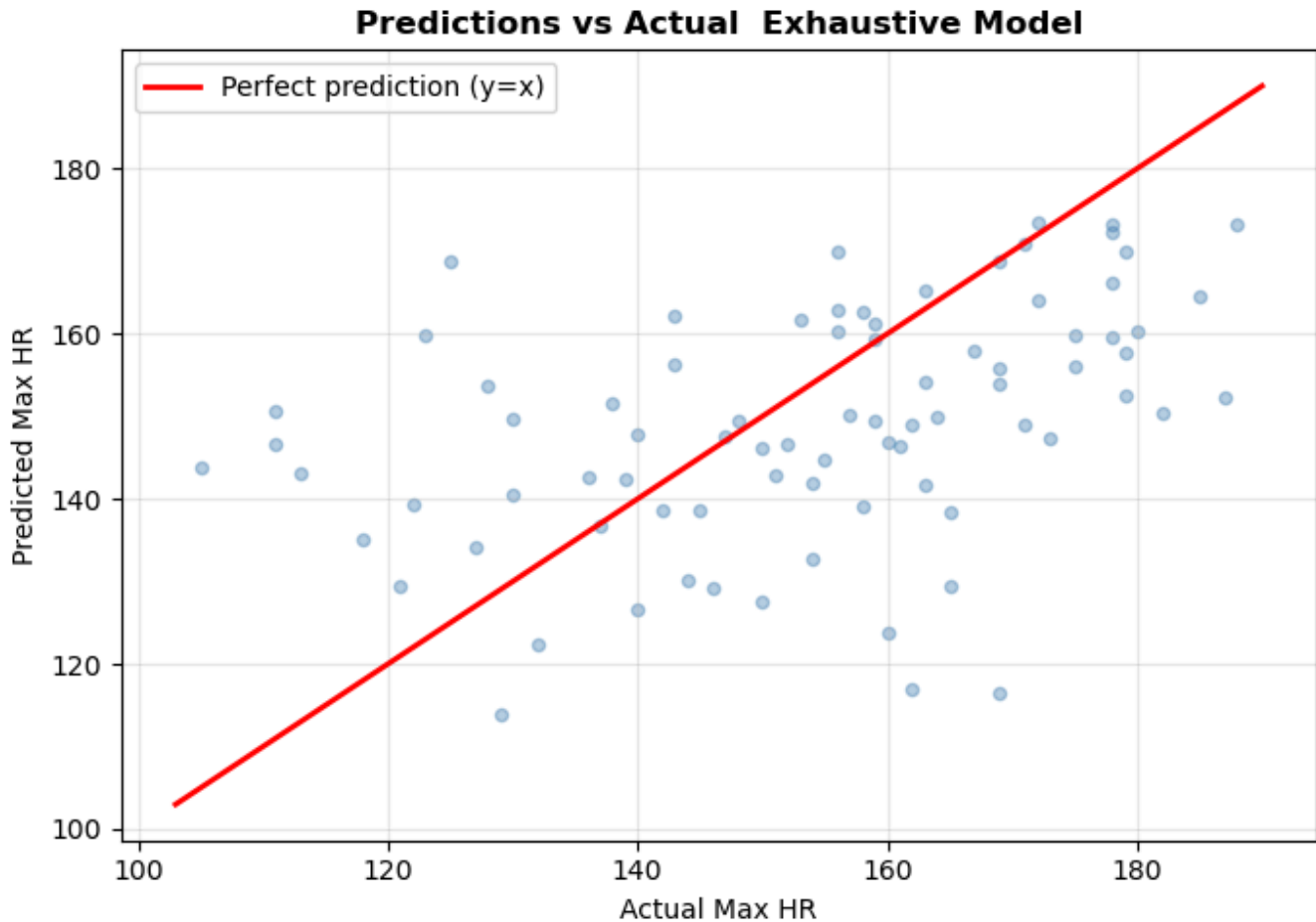
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.151e-16	0.055	-3.93e-15	1.000	-0.108	0.108
Age	-0.3194	0.062	-5.191	0.000	-0.441	-0.198
BP	0.1511	0.059	2.570	0.011	0.035	0.267
ST_dep	-0.0845	0.073	-1.153	0.250	-0.229	0.060
Ex_angina	-0.2443	0.064	-3.837	0.000	-0.370	-0.119
Chest_pain	-0.0948	0.062	-1.531	0.128	-0.217	0.027
Slope_ST	-0.2399	0.069	-3.464	0.001	-0.377	-0.103
Vessels	-0.1259	0.062	-2.040	0.043	-0.248	-0.004

Omnibus:	12.572	Durbin-Watson:	1.913
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.272
Skew:	-0.575	Prob(JB):	0.00131
Kurtosis:	3.602	Cond. No.	2.57

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
' Age      : |β|=0.3194  p=5.5703e-07  ✓'
' Ex_angina : |β|=0.2443  p=1.7203e-04  ✓'
' Slope_ST  : |β|=0.2399  p=6.6345e-04  ✓'
' BP        : |β|=0.1511  p=1.0976e-02  ✓'
' Vessels   : |β|=0.1259  p=4.2777e-02  ✓'
' Chest_pain : |β|=0.0948  p=1.2762e-01  X'
' ST_dep    : |β|=0.0845  p=2.5034e-01  X'
```



$R^2 = 0.456$, meaning the model explains only 45.6% of the variation in Max HR. Age is the most important predictor ($|\beta| = 0.319$), followed by exercise angina and slope of ST. ST depression and chest pain are not significant at $p < 0.05$. The predictions vs actual plot shows a wide scatter around the diagonal, the model captures the general trend but with considerable error. This is expected given the weak correlations with Max HR.

7. Forward Selection

```
' + Ex_angina      p=1.7648e-10  Adj R²=0.19180'
' + Age           p=1.8482e-09  Adj R²=0.33124'
' + Slope_ST      p=9.7779e-06  Adj R²=0.39524'
' + Vessels       p=1.1011e-02  Adj R²=0.41300'
' + BP           p=1.2780e-02  Adj R²=0.42950'
STOP no remaining variable has p < 0.05
```

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.445
Model:	OLS	Adj. R-squared:	0.430
Method:	Least Squares	F-statistic:	29.31
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	8.59e-22
Time:	20:11:00	Log-Likelihood:	-814.32
No. Observations:	189	AIC:	1641.
Df Residuals:	183	BIC:	1660.
Df Model:	5		
Covariance Type:	nonrobust		

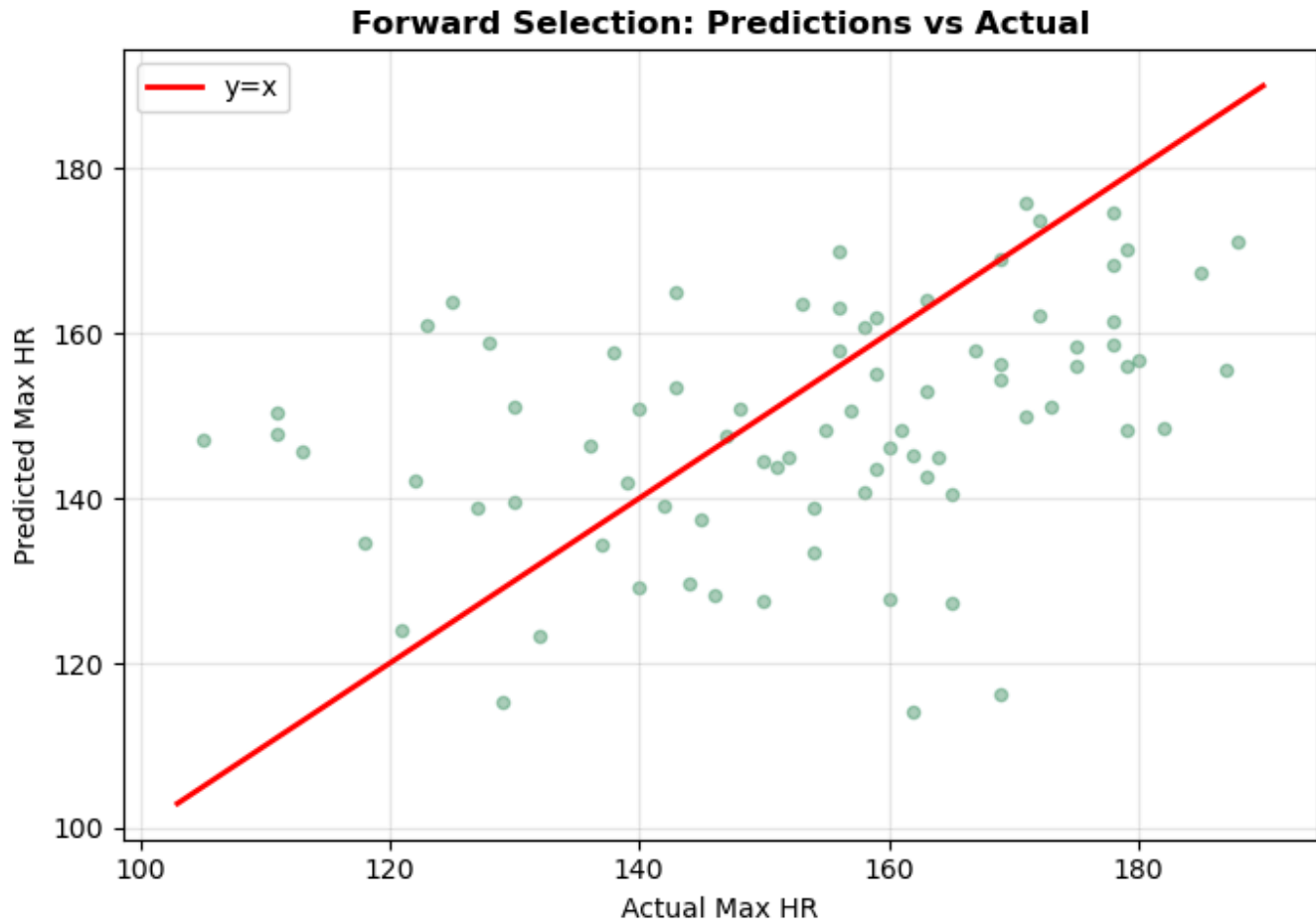
	coef	std err	t	P> t 	[0.025	0.975]
Intercept	194.7190	11.476	16.967	0.000	172.076	217.362
Ex_angina	-15.0408	3.009	-4.998	0.000	-20.978	-9.103
Age	-0.8696	0.166	-5.224	0.000	-1.198	-0.541
Slope_ST	-10.7611	2.264	-4.754	0.000	-15.227	-6.295
Vessels	-4.0359	1.503	-2.686	0.008	-7.000	-1.071
BP	0.1949	0.077	2.515	0.013	0.042	0.348

Omnibus:	12.341	Durbin-Watson:	1.860
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.229
Skew:	-0.549	Prob(JB):	0.00134
Kurtosis:	3.687	Cond. No.	1.24e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.24e+03. This might indicate that there are strong multicollinearity or other numerical problems.



Forward selection stopped at 5 variables: Ex_angina, Age, Slope_ST, Vessels, BP. ST_dep and Chest_pain were never added, their p-values exceeded 0.05 in the context of the already-selected variables. The scatter plot shows the same wide spread as the exhaustive model.

8. Backward Selection

```
- Cholesterol      p=0.3620  Adj R²=0.43485
- Thallium        p=0.3388  Adj R²=0.43510
- ST_dep          p=0.2503  Adj R²=0.43408
- Chest_pain      p=0.1171  Adj R²=0.42950
STOP all remaining variables have p < 0.05
```

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.445
Model:	OLS	Adj. R-squared:	0.430
Method:	Least Squares	F-statistic:	29.31
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	8.59e-22
Time:	20:11:22	Log-Likelihood:	-814.32
No. Observations:	189	AIC:	1641.
Df Residuals:	183	BIC:	1660.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
Intercept	194.7190	11.476	16.967	0.000	172.076	217.362
Age	-0.8696	0.166	-5.224	0.000	-1.198	-0.541
BP	0.1949	0.077	2.515	0.013	0.042	0.348
Ex_angina	-15.0408	3.009	-4.998	0.000	-20.978	-9.103
Slope_ST	-10.7611	2.264	-4.754	0.000	-15.227	-6.295
Vessels	-4.0359	1.503	-2.686	0.008	-7.000	-1.071

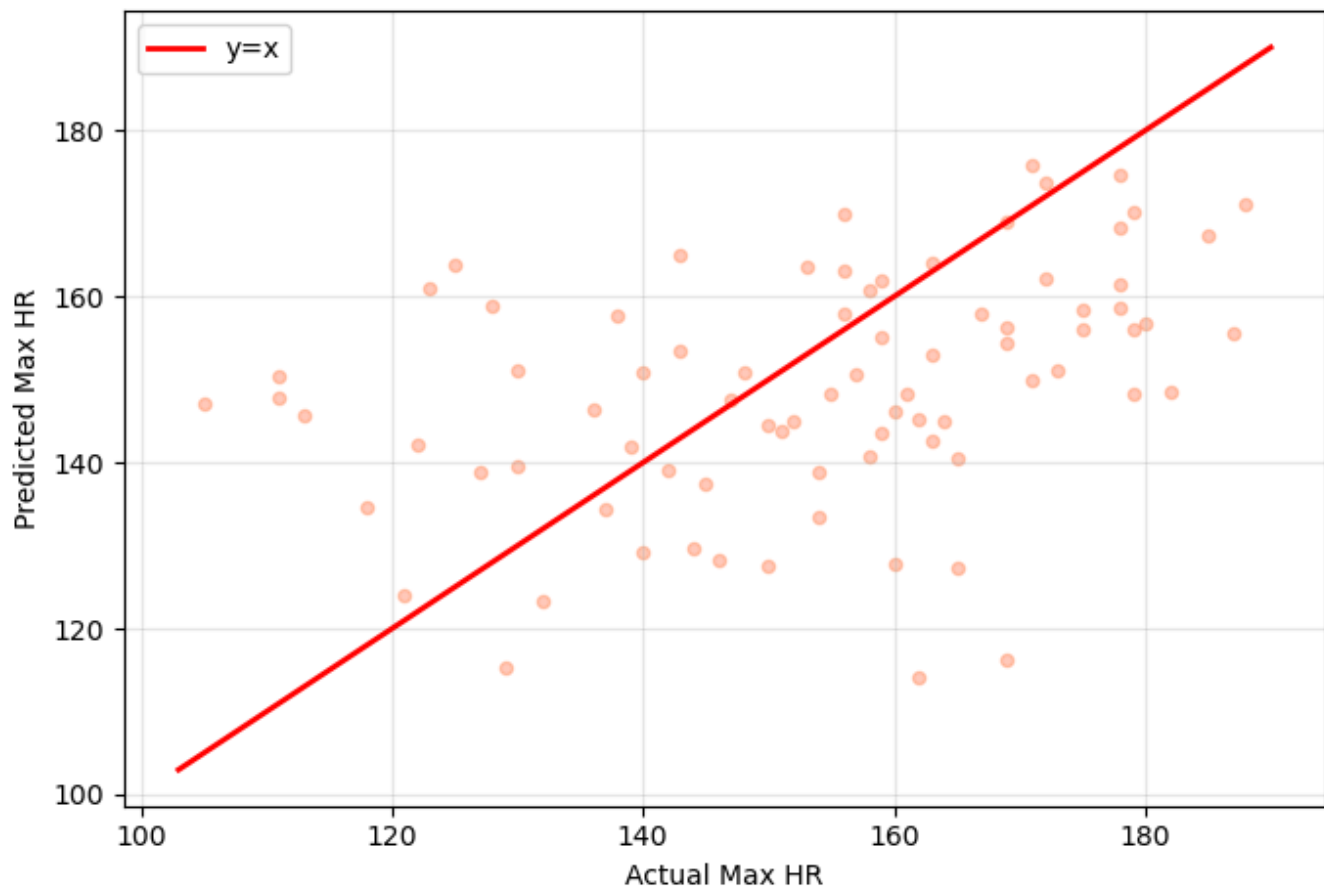
Omnibus:	12.341	Durbin-Watson:	1.860
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.229
Skew:	-0.549	Prob(JB):	0.00134
Kurtosis:	3.687	Cond. No.	1.24e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.24e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Backward Selection: Predictions vs Actual

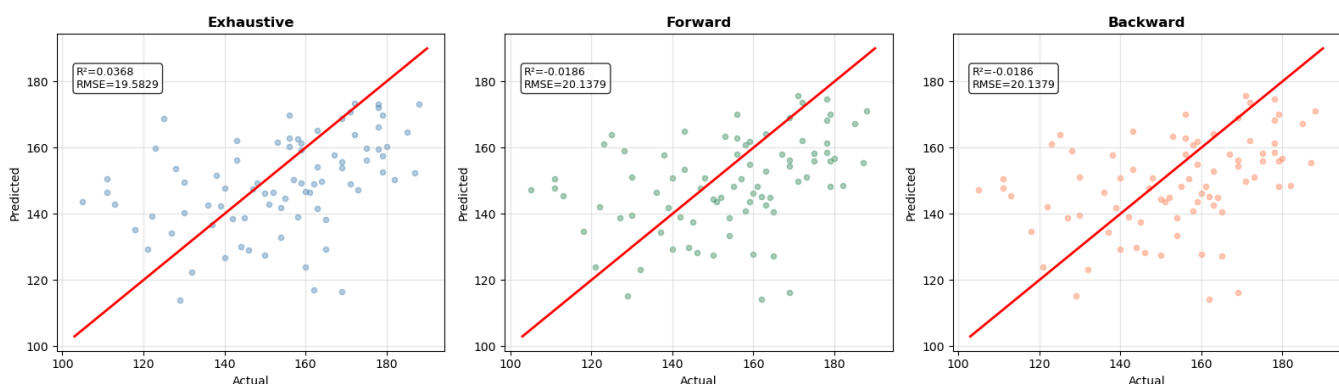


Backward selection removed Cholesterol, Thallium, ST_dep, and Chest_pain sequentially. Final selected variables: Age, BP, Ex_angina, Slope_ST, Vessels, identical to forward selection. Both methods converge on the same 5 variables.

9. Baseline Performance : Test Set

Model	RMSE	R2
Exhaustive	19.5829	0.0368
Forward	20.1379	-0.0186
Backward	20.1379	-0.0186

Comparison: Predictions vs Actual



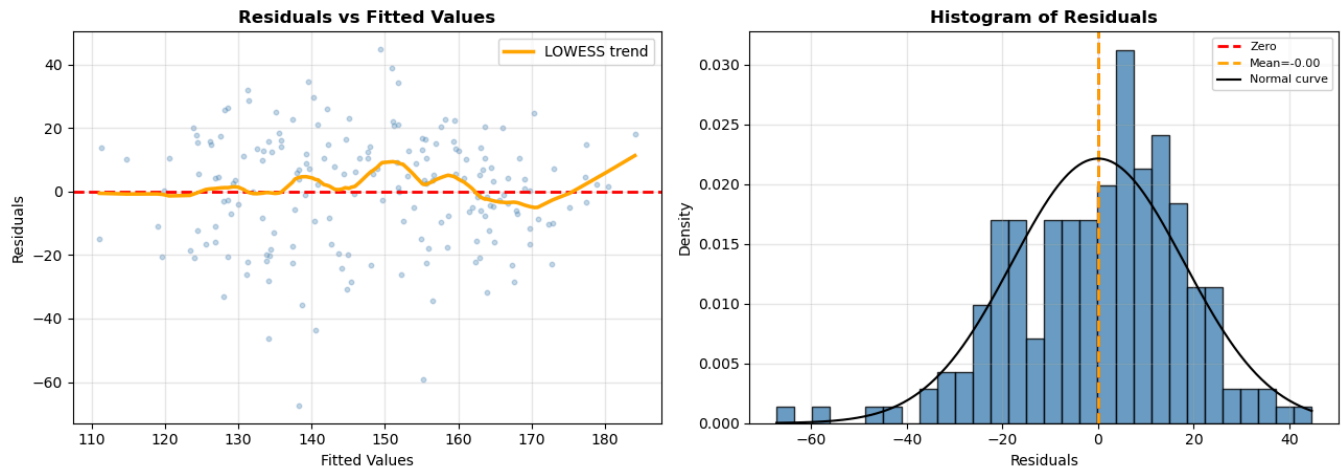
The exhaustive model achieves the best test R^2 (0.037) but this is essentially zero, all three models fail to generalise to unseen patients. Forward and backward produce negative R^2 (-0.019), meaning they perform worse than simply predicting the mean Max HR for everyone. This is a strong signal that Max HR cannot be reliably predicted from these variables using linear regression on only 270 patients.

10. Residual Diagnostics

Mean of residuals : -2.22e-13

Median of residuals : 2.6907

Residual Diagnostics



Mean of residuals is effectively zero as guaranteed by OLS. The histogram is roughly bell-shaped but with a slight left skew, the distribution is not perfectly normal. The residuals vs fitted plot shows a relatively random scatter with no strong pattern, suggesting homoscedasticity is approximately satisfied. The LOWESS line is mostly flat with some noise due to the small sample size.

11. Polynomial Regression: Adding Age²

Age is the dominant predictor. We test whether the relationship between age and Max HR is linear or curved. The Hierarchy Principle requires keeping Age whenever Age² is included.

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.459
Model:	OLS	Adj. R-squared:	0.435
Method:	Least Squares	F-statistic:	19.08
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	1.31e-20
Time:	20:11:49	Log-Likelihood:	-811.88
No. Observations:	189	AIC:	1642.
Df Residuals:	180	BIC:	1671.
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	161.8759	41.374	3.913	0.000	80.236	243.516
Age	0.5898	1.546	0.381	0.703	-2.461	3.641
BP	0.1991	0.079	2.530	0.012	0.044	0.354
ST_dep	-1.7924	1.512	-1.185	0.237	-4.776	1.192
Ex_angina	-12.8924	3.294	-3.914	0.000	-19.392	-6.393
Chest_pain	-2.4672	1.568	-1.574	0.117	-5.560	0.626
Slope_ST	-9.1571	2.672	-3.427	0.001	-14.430	-3.884
Vessels	-3.1920	1.560	-2.046	0.042	-6.270	-0.114
Age_sq	-0.0136	0.014	-0.943	0.347	-0.042	0.015

Omnibus:	12.691	Durbin-Watson:	1.944
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.437
Skew:	-0.577	Prob(JB):	0.00121
Kurtosis:	3.613	Cond. No.	9.98e+04

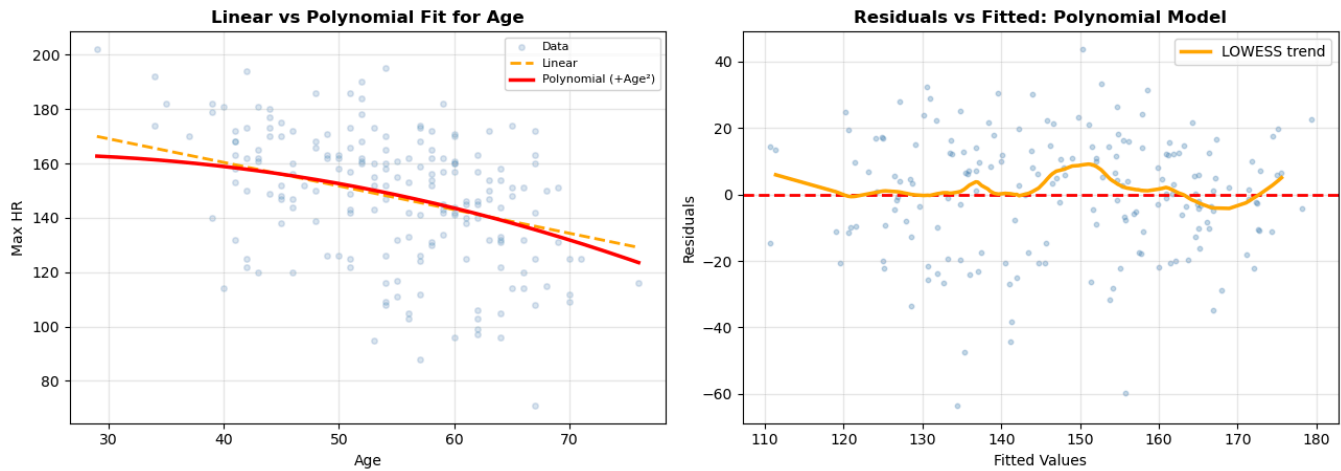
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 9.98e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Age^2 : $\beta = -1.355688\text{e-}02$ $p = 3.4677\text{e-}01$
 Backward Adj R^2 : 0.42950
 Polynomial Adj R^2 : 0.43476
 Gain : +0.526 pp

Polynomial Regression: Age + Age²



Age² is not significant ($p = 0.347$). The polynomial term adds almost nothing, gain of only +0.5 pp. The left panel shows the linear and polynomial lines nearly overlapping, confirming that the relationship between age and Max HR is already approximately linear in this dataset. Unlike the CASP dataset where F2² gave +10 pp, the polynomial does not help here.

12. Interaction Model: Age × ST Depression

We test whether the effect of age on Max HR depends on the level of ST depression. whether older patients with higher ST depression have disproportionately lower Max HR than the sum of the individual effects.

OLS Regression Results

Dep. Variable:	Max_HR	R-squared:	0.458
Model:	OLS	Adj. R-squared:	0.434
Method:	Least Squares	F-statistic:	19.03
Date:	Thu, 19 Mar 2026	Prob (F-statistic):	1.45e-20
Time:	20:15:05	Log-Likelihood:	-811.99
No. Observations:	189	AIC:	1642.
Df Residuals:	180	BIC:	1671.
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	204.1725	14.485	14.095	0.000	175.590	232.755
Age	-0.9709	0.213	-4.549	0.000	-1.392	-0.550
ST_dep	-9.1067	9.087	-1.002	0.318	-27.037	8.823
Age:ST_dep	0.1316	0.160	0.822	0.412	-0.184	0.448
BP	0.2037	0.079	2.588	0.010	0.048	0.359
Ex_angina	-12.2510	3.299	-3.714	0.000	-18.760	-5.742
Chest_pain	-2.2185	1.581	-1.403	0.162	-5.339	0.902
Slope_ST	-9.3395	2.674	-3.492	0.001	-14.617	-4.062
Vessels	-3.3697	1.577	-2.136	0.034	-6.482	-0.257

Omnibus:	12.088	Durbin-Watson:	1.913
Prob(Omnibus):	0.002	Jarque-Bera (JB):	12.691
Skew:	-0.559	Prob(JB):	0.00175
Kurtosis:	3.600	Cond. No.	1.83e+03

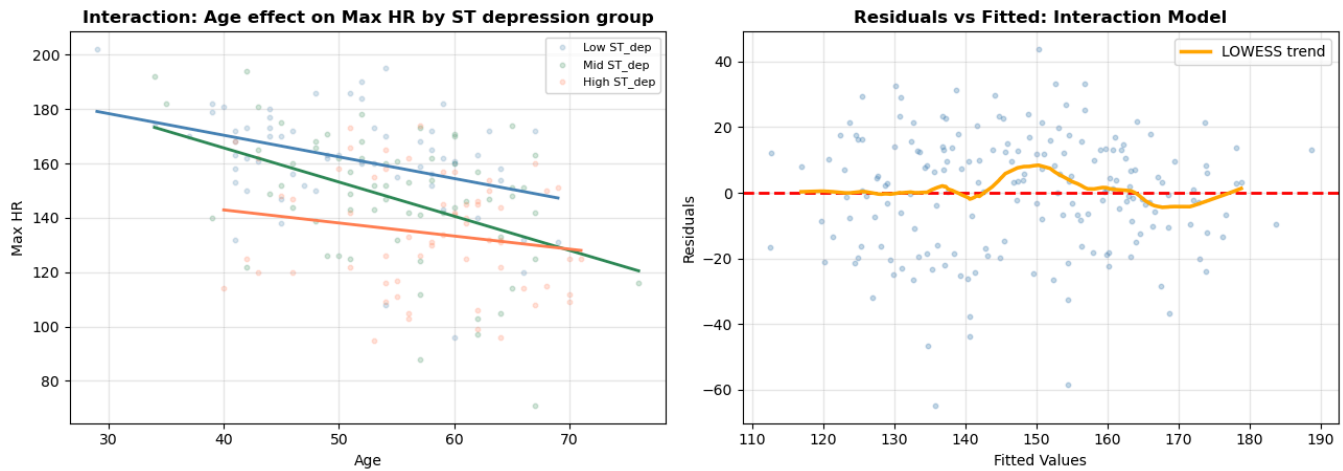
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.83e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Age $\times$ ST_dep: $\beta = 1.316494\text{e-}01$ $p = 4.1221\text{e-}01$
 Backward Adj R^2 : 0.42950
 Interaction Adj R^2 : 0.43409
 Gain : +0.459 pp

Interaction Model: Age $\times$ ST depression

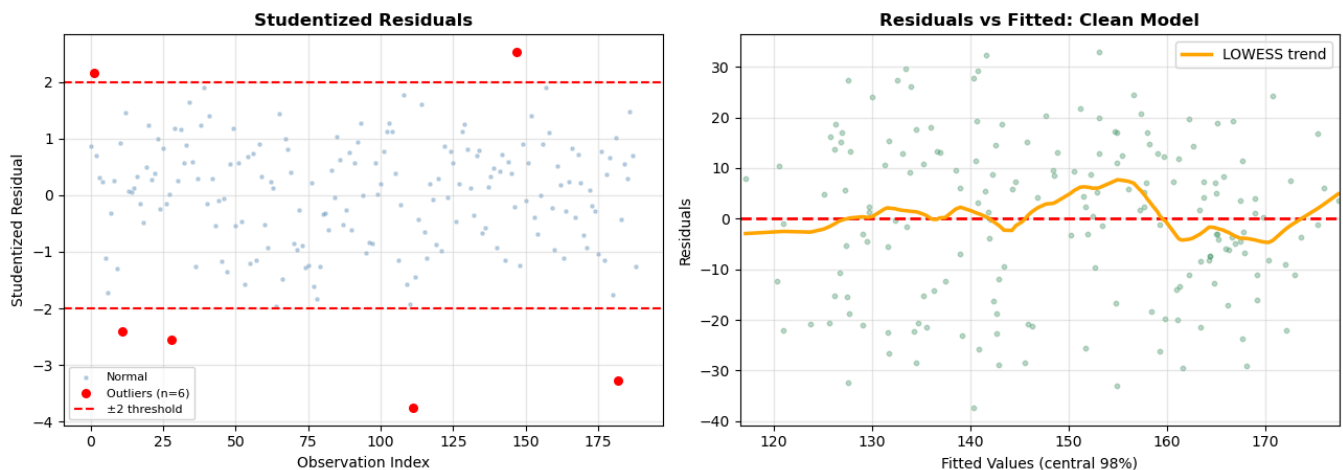


The interaction term Age:ST_dep is not significant ($p = 0.412$). The three ST depression groups show similar slopes in the left panel, confirming the effect of age on Max HR does not meaningfully depend on ST depression level. Adj R^2 gain is only +0.46 pp. Unlike the CASP interaction, this one adds no real value.

13. Outlier Detection and Clean Model

Training patients : 189
 Flagged ($|e^*| > 2$) : 6 (3.2%)
 Clean training set : 183 patients

Outlier Detection and Clean Model



Backward model – Test R^2 : -0.0186 RMSE: 20.1379
 Clean model – Test R^2 : -0.0089 RMSE: 20.0418
 Train R^2 original : 0.44467
 Train R^2 clean : 0.50786

Only 6 outliers flagged (3.2%), fewer than expected from a normal distribution. The left panel shows most residuals within the ± 2 threshold. Removing outliers improves training R^2 from 0.445 to 0.508 but test R^2

remains near zero (-0.009). Removing outliers from an already poor model does not fix the underlying problem, the predictors simply do not explain enough of the variance in Max HR.

14. Final Model Comparison

Model	Train Adj R ²	Test R ²	Test RMSE
Exhaustive	0.43510	0.0368	19.5829
Forward	0.42950	-0.0186	20.1379
Backward	0.42950	-0.0186	20.1379
Polynomial + Age ²	0.43476	0.0033	19.9206
Interaction Age×ST_dep	0.43409	0.0081	19.8722
Clean Backward	0.49395	-0.0089	20.0418

Best model: Exhaustive
 Test R² = 0.03680
 Test RMSE = 19.5829

All six models produce near-zero or negative test R², confirming that Max HR cannot be reliably predicted from these variables using linear regression on 270 patients. The exhaustive model technically wins with test R² = 0.037 but this is not a meaningful result. The polynomial and interaction terms, which produced large improvements in the CASP dataset, add almost nothing here, confirming the problem is not the model form but the weakness of the signal itself.

Conclusion

In this analysis we applied the full regression pipeline to predict maximum heart rate from clinical and demographic measurements on 270 patients.

We found that all six models, exhaustive, forward, backward, polynomial, interaction, and clean, produced near-zero or negative test R². The best result we achieved was a test R² of 0.037 from the exhaustive model, which is not a meaningful predictive performance.

We identified age as the strongest predictor ($r = -0.40$), followed by exercise angina and slope of ST. However, even combining all significant predictors was not sufficient to build a reliable model. Adding Age² and an Age × ST depression interaction, approaches that produced large improvements in the CASP dataset, added almost nothing here.

We concluded that this failure is not a modelling error but a data problem. The signal between these cardiac measurements and maximum heart rate is simply too weak, and 270 patients is too small a sample for linear regression to learn stable generalizable patterns. This is itself a scientific finding, it tells us that maximum heart rate is likely driven by factors not captured in this dataset, such as fitness level, medication, or genetic predisposition.